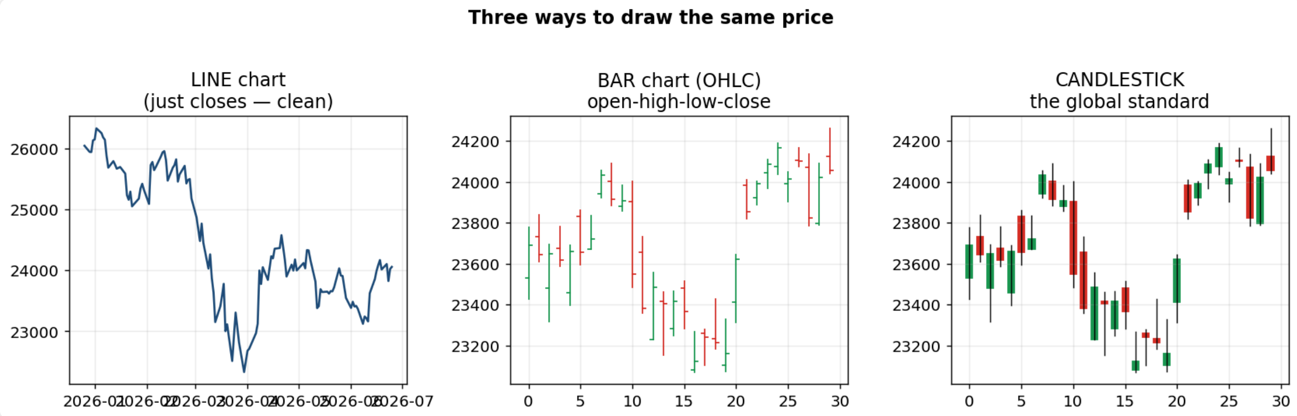
# Technical Analysis
# Illustrated Handbook

Every topic: the chart, why it works, and the market psychology behind it.
Built to answer the interviewer's follow-up question.

Prepared for Saikumar Kaleru — Technical Research Analyst prep

# PART 1 — Foundations

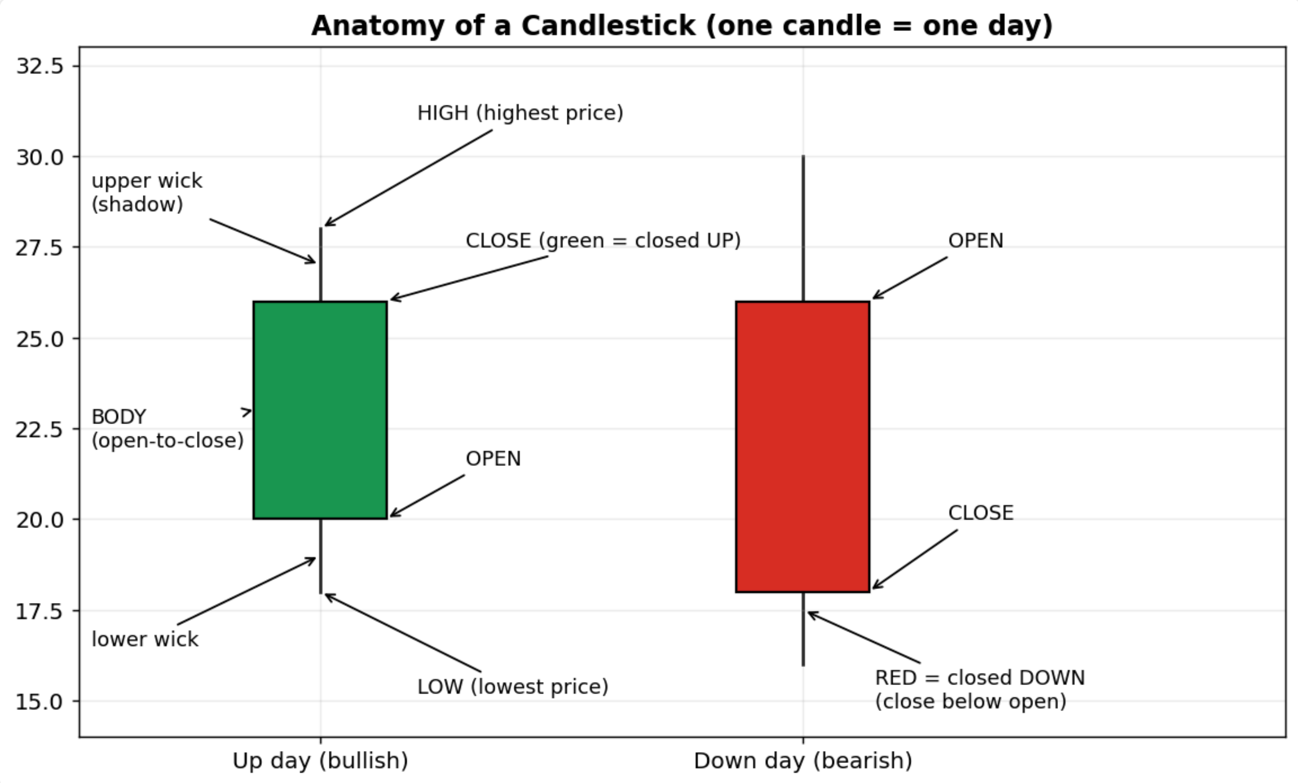
## Three ways to draw price




**What it is:** The same price history shown as a line (closes only), a bar (OHLC), or a candlestick chart.
**Why it works:** Candlesticks encode four prices (open, high, low, close) plus the winner of the day in one visual symbol, so you read the balance of power at a glance — that's why they became the global standard.
**The psychology:** Each candle is a snapshot of a battle between buyers and sellers. A big body = one side dominated; small body or long wicks = conflict and indecision. Charts work because they are a picture of collective human emotion, and emotion repeats.
**How to use / trade it:** Use candlesticks as your default; drop to a line chart when you just want to see the clean trend without noise.

> **Every part, explained simply:** Same prices, three views. **Line** connects only the closes — cleanest for spotting the big trend. **Bar (OHLC)** shows open-high-low-close as a tick chart. **Candlestick** shows the same four prices but colour-coded so you instantly see who won the day — the global standard.

## What one candle tells you




**What it is:** Body = open-to-close range; wicks/shadows = the high and low; green = closed up, red = closed down.

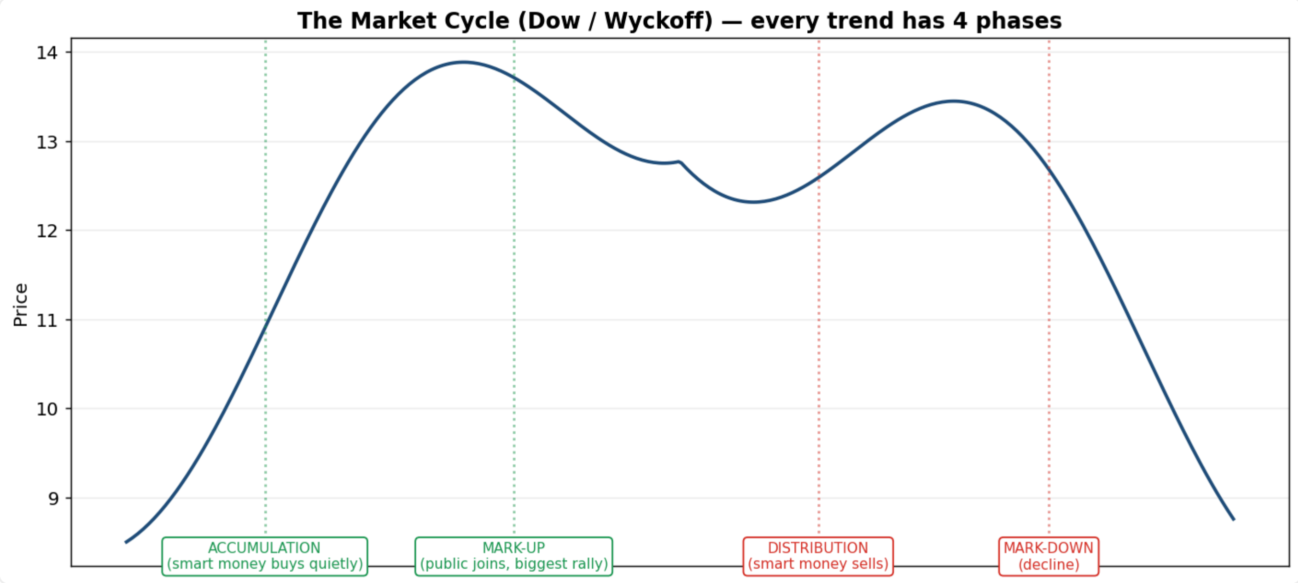
**Why it works:** The body shows who won and by how much; the wicks show where price was rejected. A long upper wick means buyers pushed up but sellers slammed it back (supply); a long lower wick means sellers pushed down but buyers absorbed it (demand).

**The psychology:** A long body reflects conviction — one side overpowered the other all session. A tiny body reflects a stalemate. Wicks are 'rejection': the market visited a price and refused to stay there, which tells you where strong orders sit.

**How to use / trade it:** Read the wick direction for hidden supply/demand: repeated long lower wicks at a level = buyers defending it (likely support).

> **Every part, explained simply:** One candle = one day (or week/month). It packs four prices: **Open** (first price), **Close** (last price), **High** (highest), **Low** (lowest). The thick **body** runs from open to close; if the close is higher than the open it's **green** (buyers won), else **red** (sellers won). The thin **wicks** show the high and low — a long wick means price went there but was pushed back.

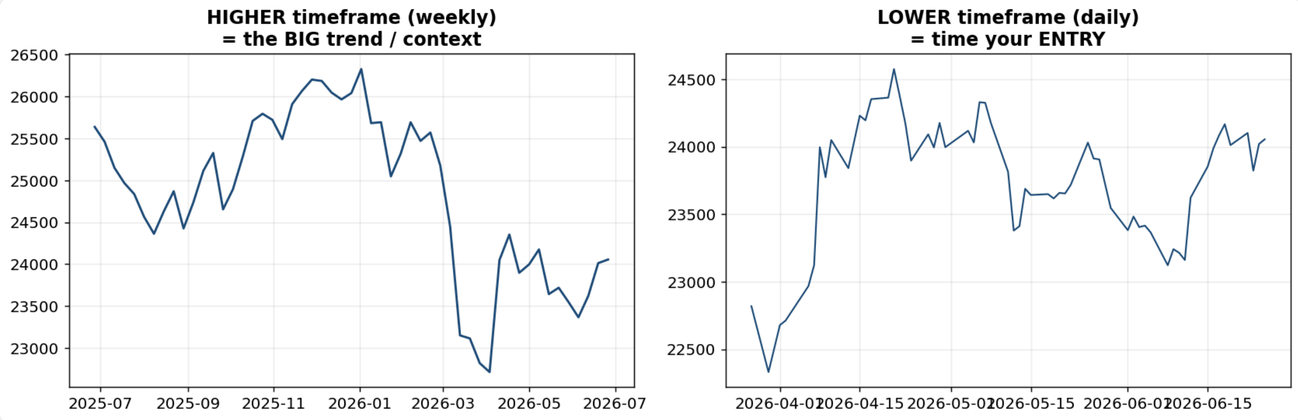
## The market cycle (Dow / Wyckoff)




**What it is:** Every trend moves through four phases: Accumulation, Mark-up, Distribution, Mark-down.
**Why it works:** Markets are driven by 'smart money' (institutions) who must buy and sell in size without moving price against themselves — so they accumulate quietly at bottoms and distribute quietly at tops, creating the repeating cycle.
**The psychology:** Accumulation happens during maximum pessimism (everyone has given up selling). Mark-up is when the public notices and FOMO drives the biggest rally. Distribution is peak euphoria, where smart money sells to the late, greedy crowd. Mark-down is the painful unwinding. It's fear → greed → fear.
**How to use / trade it:** Identify the phase from price+volume: sideways ranges after a big fall (low volume) often = accumulation; sideways after a big rally with high churn = distribution.
**Interview depth:** This is the backbone of Dow Theory and Wyckoff — name it and explain that smart money buys when retail panics and sells when retail is euphoric.

> **Every part, explained simply:** Every trend has four phases. **Accumulation**: smart money quietly buys at the bottom while the public is fearful (sideways, low volume). **Mark-up**: the public notices and piles in — the biggest rally. **Distribution**: smart money quietly sells to the greedy crowd at the top. **Mark-down**: the decline. Knowing the phase tells you whether to buy or be cautious.

## Multiple-timeframe analysis




**What it is:** Reading the same asset on a higher timeframe (weekly) and a lower one (daily) together.

**Why it works:** A signal aligned with the bigger trend has the wind at its back; a signal against it is fighting a stronger force and usually fails.

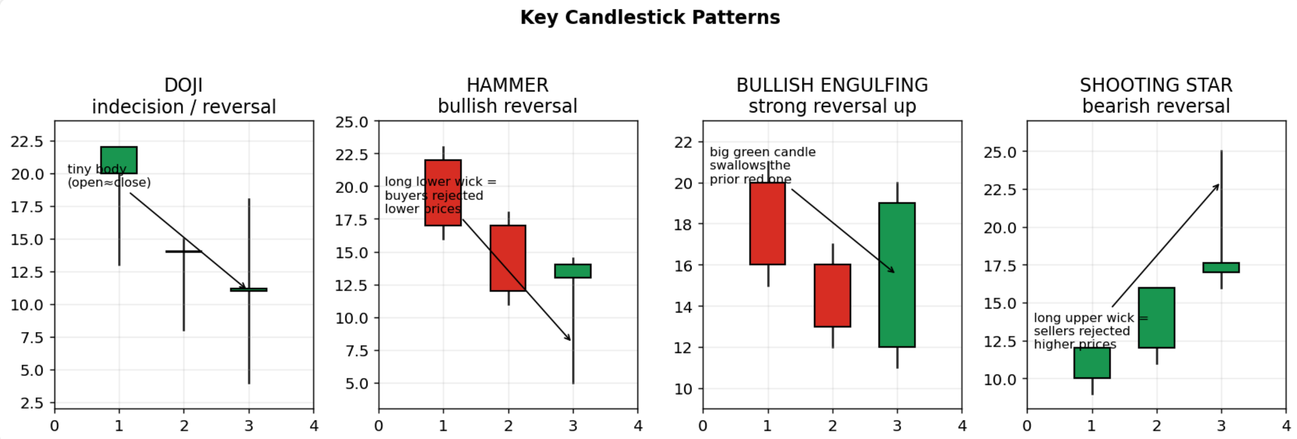
**The psychology:** Different participants act on different timeframes — long-term investors set the tide, short-term traders create the ripples. When both agree, the crowd is unified and moves are powerful.

**How to use / trade it:** Decide direction on the higher timeframe (only take longs if the weekly trend is up), then time the entry on the lower one. Conflicting timeframes = lower conviction, smaller size or skip.

> **Every part, explained simply:** Always read two timeframes. The **higher** one (e.g. weekly) shows the dominant trend — your context and bias. The **lower** one (e.g. daily/hourly) is where you time the entry. Rule: only trade in the direction of the higher timeframe; if they conflict, it's a low-conviction trade.

# PART 2 — Price Action & Chart Patterns

## Reversal candlestick patterns




**What it is:** Doji (indecision), Hammer (bullish reversal), Bullish Engulfing (strong reversal up), Shooting Star (bearish reversal).

**Why it works:** Each captures a sudden shift in the buyer/seller balance at the end of a move, which often precedes a turn.

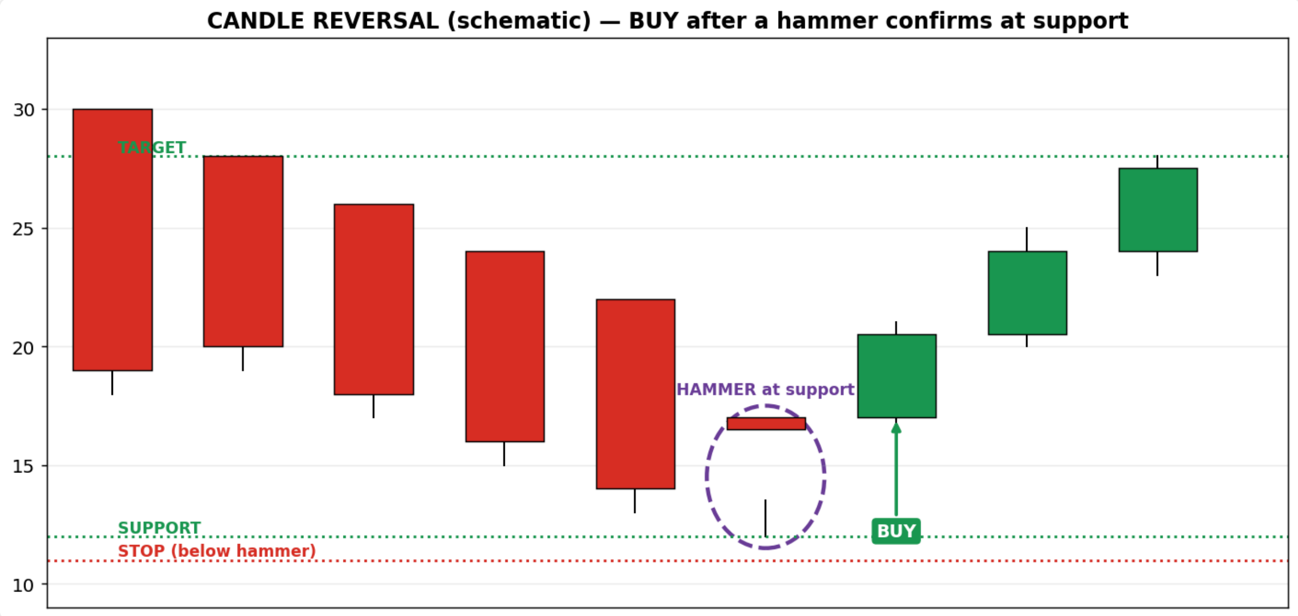
**The psychology: Doji**: after a strong trend, buyers and sellers reach equilibrium — the side that was winning has lost control. **Hammer**: in a downtrend, sellers drive price far down but buyers storm in and close near the high — sellers are now trapped and must cover, fuelling a bounce. **Bullish engulfing**: sellers open confidently lower, then buyers overwhelm them and close above the entire prior candle — a visible sentiment flip that traps shorts. **Shooting star**: buyers make a new high but sellers reject it hard — demand is exhausted.

**How to use / trade it:** Only act on these AT a level (support/resistance/MA) and ideally with above-average volume — that confluence is what makes them reliable rather than random.

**Interview depth:** Stress that a candle in mid-air means little; the SAME candle at a major support with high volume is a high-probability signal because trapped traders are forced to act.

> **Every part, explained simply:** Four classic reversal candles:
>
> - **Doji** — open and close almost equal (tiny body) = buyers and sellers balanced = indecision, often before a turn.
> - **Hammer** — small body on top, long lower wick; after a fall it shows buyers slammed price back up = bullish.
> - **Bullish Engulfing** — a big green candle completely covers the previous red one = buyers overwhelmed sellers = strong reversal up.
> - **Shooting Star** — small body at the bottom, long upper wick; after a rise it shows sellers rejected higher prices = bearish.
>
> They matter most AT a support/resistance level, on above-average volume.

## Trading a reversal — the hammer buy (worked example)

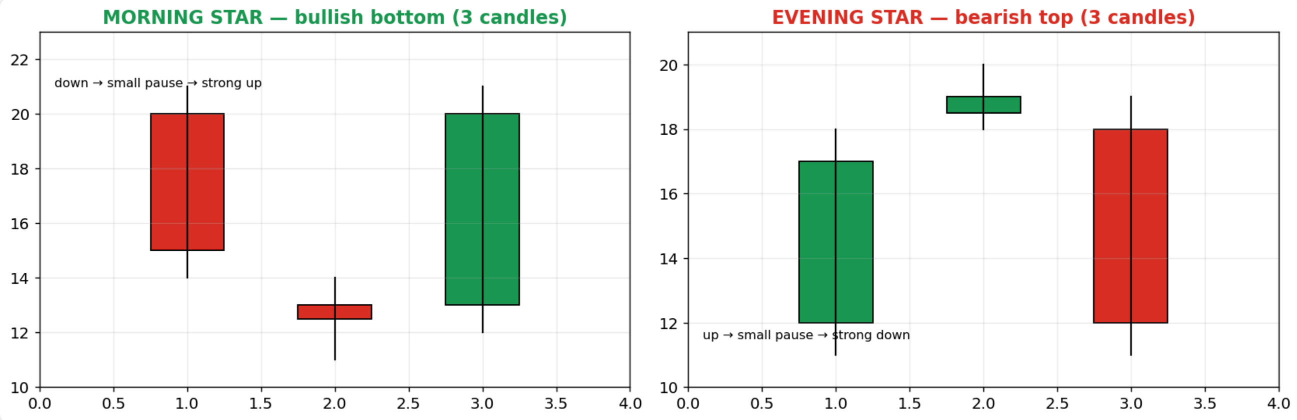



**What it is:** A step-by-step example of trading ONE reversal candle: a hammer that appears at support.
**Why it works:** The hammer shows sellers tried to push lower but buyers won the day back — the first sign of a turn.
**The psychology:** Sellers who shorted the lows are now trapped; as price rises they're forced to buy back, adding fuel.
**How to use / trade it:** Wait for the NEXT candle to close above the hammer's high before buying — that confirms it; stop just below the hammer, target the next resistance.
**Interview depth:** Never trade a hammer floating in mid-air — it only counts at a level like support.

> **Every part, explained simply:** A **hammer** has a small body at the top and a long lower wick. The story: sellers shoved price far down during the day (long tail), but buyers fought it all the way back up by the close — so buyers are taking over. You don't buy the hammer itself; you wait for the NEXT candle to close higher, which confirms the reversal. Strongest when it appears right at a support level.

> **Trade plan — where to buy, stop and sell:**
> - **Buy / entry:** After a bullish reversal candle (hammer / bullish engulfing) forms AT a support level, buy when the NEXT candle closes above the pattern's high (that confirms buyers followed through).
> - **Stop-loss:** Just below the pattern's low (under the wick) — if price goes there, the reversal failed.
> - **Target / when to sell:** The next resistance level above, sized for at least 2:1 reward-to-risk. Sell there, or trail your stop up under each higher low.

## Three-candle star reversals




**What it is:** Morning Star (bullish bottom) and Evening Star (bearish top) — a trend candle, a small 'star' of indecision, then a strong candle the other way.

**Why it works:** They show a three-act transfer of control: dominance, hesitation, then reversal — more reliable than a single candle because the shift is confirmed over three sessions.

**The psychology:** Act 1: the trend looks healthy. Act 2: the small star shows the dominant side suddenly can't push further (doubt creeps in). Act 3: the opposite side seizes control with conviction. The pause is the moment the crowd's belief cracks.

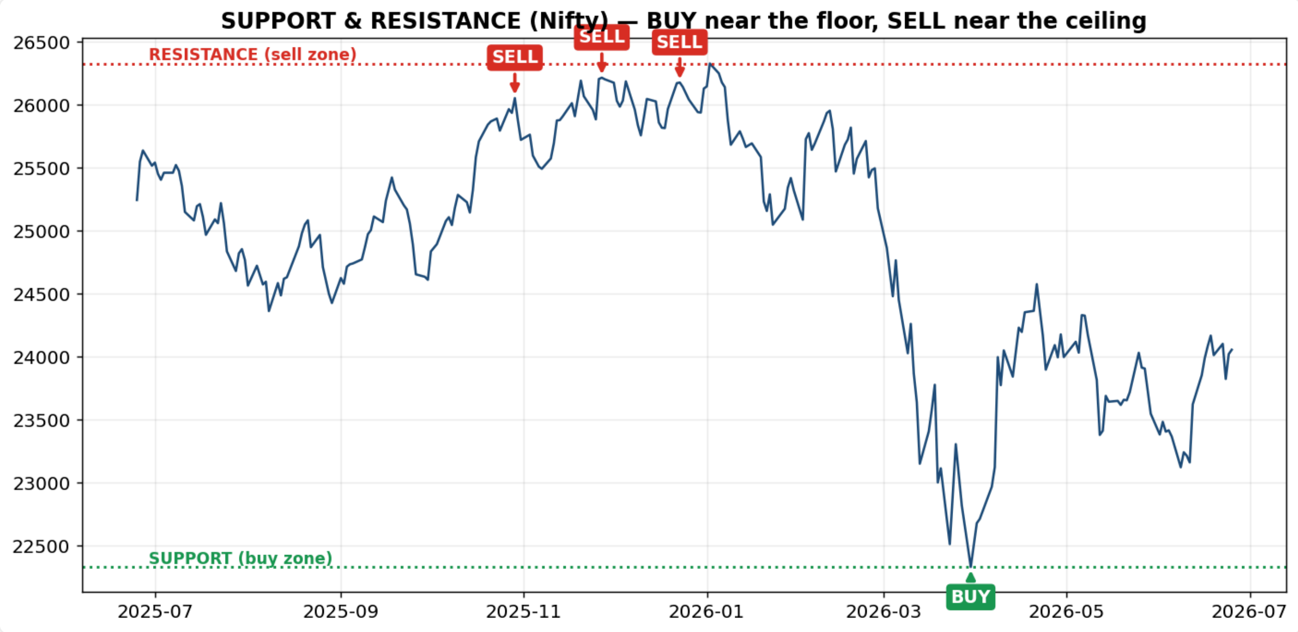
**How to use / trade it:** Enter on the third candle's confirmation; place the stop beyond the star's extreme.

> **Every part, explained simply:** Both are three-candle reversals. **Morning Star** (bullish): a big down candle, then a small 'star' candle (indecision = the dominant sellers lose steam), then a big up candle = bottom. **Evening Star** (bearish): big up candle, small star, big down candle = top. The little middle candle is the moment the crowd's belief cracks.

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** Morning Star: buy when the 3rd (strong green) candle closes. Evening Star: sell/exit when the 3rd red candle closes.
- **Stop-loss:** Just beyond the middle 'star' candle's extreme (below the low for a Morning Star).
- **Target / when to sell:** The prior swing high (Morning Star) or swing low (Evening Star); aim for 2:1 or better.

## Support & resistance — the floor and the ceiling




**What it is:** Support = a price floor where buyers repeatedly step in; resistance = a ceiling where sellers repeatedly step in.

**Why it works:** They work because of MEMORY and order clustering: traders remember prices where reversals happened and place orders there again, which makes the level hold — partly self-fulfilling.

**The psychology:** At support three groups buy at once: those who missed the last rally and want a second chance, those who bought there before and add, and shorts taking profit. That clustered demand stops the fall. **Role reversal**: when resistance finally breaks, traders who sold there are now trapped at a loss and buy back at breakeven on the retest — turning old resistance into new support.

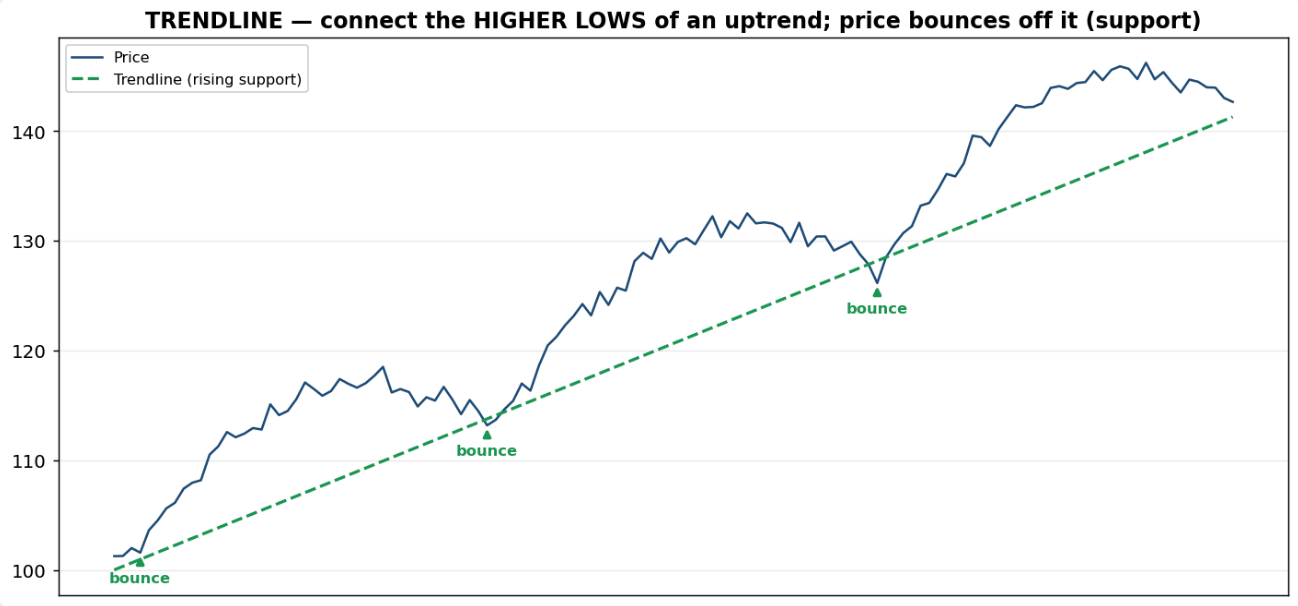
**How to use / trade it:** Buy near support / sell near resistance with a stop just beyond the level; trade the breakout when price clears it on strong volume.

**Interview depth:** The more times a level is tested the more significant it is — but each test consumes orders, so a level tested many times eventually breaks (the supply/demand there gets used up).

> **Every part, explained simply: Support** is a price floor where buyers keep stepping in and stop the fall. **Resistance** is a ceiling where sellers keep appearing and stop the rise. They exist because markets have memory — traders remember those levels and act there again. When price finally breaks through a ceiling, that old resistance often becomes the new support (doubters now buy the dip back to it).

> **Trade plan — where to buy, stop and sell:**
> • **Buy / entry:** Near SUPPORT, buy only once a bullish candle shows the level is holding (don't catch a falling knife). For a breakout, buy when a candle CLOSES above resistance — ideally on the retest from above.
> • **Stop-loss:** Just below support (or below the broken resistance on a breakout trade).
> • **Target / when to sell:** The next resistance above. For a breakout, add the range's height to the breakout point. Sell into resistance or trail the stop.

## Trendlines




**What it is:** A line connecting the rising lows of an uptrend (or falling highs of a downtrend); a sloping support/resistance.

**Why it works:** It visualises that buyers are consistently willing to step in at HIGHER prices each time — proof of increasing demand.

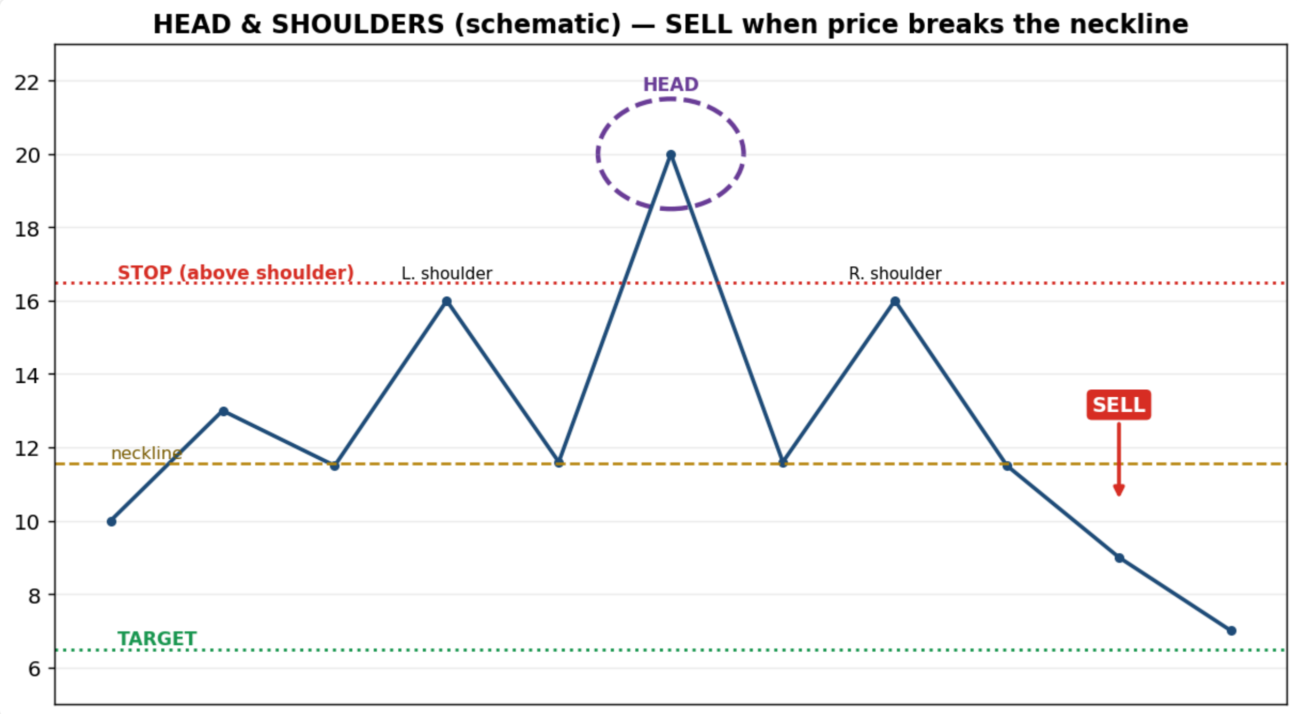
**The psychology:** Each bounce off the line is buyers showing growing optimism (paying up earlier each dip). A break of the line means that rising demand has finally failed — the psychological 'agreement' to keep buying higher has broken.

**How to use / trade it:** Needs two touches to draw, a third to confirm. Buy bounces off the line in an uptrend; treat a decisive break as an early warning to tighten stops.

> **Every part, explained simply:** A trendline is a straight line drawn under the rising lows of an uptrend (or over the falling highs of a downtrend). It's a sloping version of support: each time price dips to the line, buyers step in. You need two touches to draw it and a third to trust it. A clean break of the line is an early warning the trend is changing.

> **Trade plan — where to buy, stop and sell:**
> - **Buy / entry:** Buy when price bounces off the rising trendline and prints a bullish candle.
> - **Stop-loss:** Just below the trendline / the bounce low.
> - **Target / when to sell:** The prior swing high; trail the stop up the line. EXIT if price closes decisively below the trendline.

## Head & Shoulders (top reversal)




**What it is:** Three peaks — a higher middle peak (head) between two lower peaks (shoulders) — with a 'neckline' support; a break below it signals a top.

**Why it works:** It maps a trend visibly running out of buyers: each rally attempt has less force, and the final support break unleashes trapped longs.

**The psychology:** Left shoulder = a normal rally with healthy volume. Head = a euphoric push to a new high but usually on LOWER volume — the last greedy buyers. Right shoulder = buyers try again but can't even make a new high — demand is exhausted. When the neckline breaks, everyone who bought the top is underwater and sells, accelerating the fall.

**How to use / trade it:** Sell on the neckline break (ideally a retest of it from below); target ≈ the head-to-neckline distance projected down. Volume should fall through the head and spike on the break.

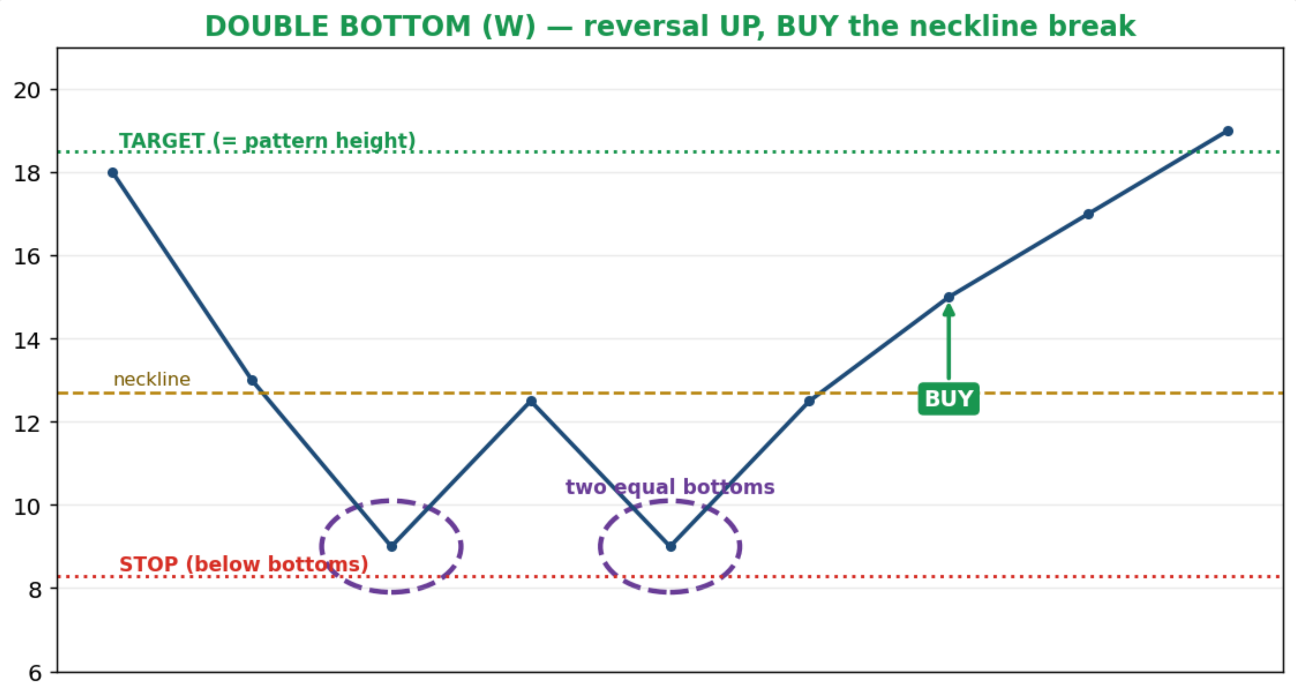
**Interview depth:** The volume signature is the tell — point out that lower volume on the head vs the left shoulder reveals weakening conviction even before the break.

> **Every part, explained simply:** Picture a silhouette: left shoulder, higher head, right shoulder. The market rallies (left shoulder), pulls back, rallies to a NEW high (head), pulls back, then only reaches a LOWER high (right shoulder) — buyers are weakening. The **neckline** is the support under the two pullback lows; a break below it starts the fall. Target ≈ the height from head to neckline.

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** This is a TOP pattern → SELL / exit longs (or short) when price closes below the neckline, or on a pullback that retests it. (Inverse H&S: do the opposite — buy the upward neckline break.)
- **Stop-loss:** Above the right shoulder.
- **Target / when to sell:** Project the head-to-neckline distance downward from the break — cover/target there.

## Double Bottom (W) — reversal up




**What it is:** Price falls to a level, bounces, falls to the SAME level again, then bounces — a 'W' shape.

**Why it works:** A level that holds twice proves strong buying support sits there; the second hold convinces sellers to give up.

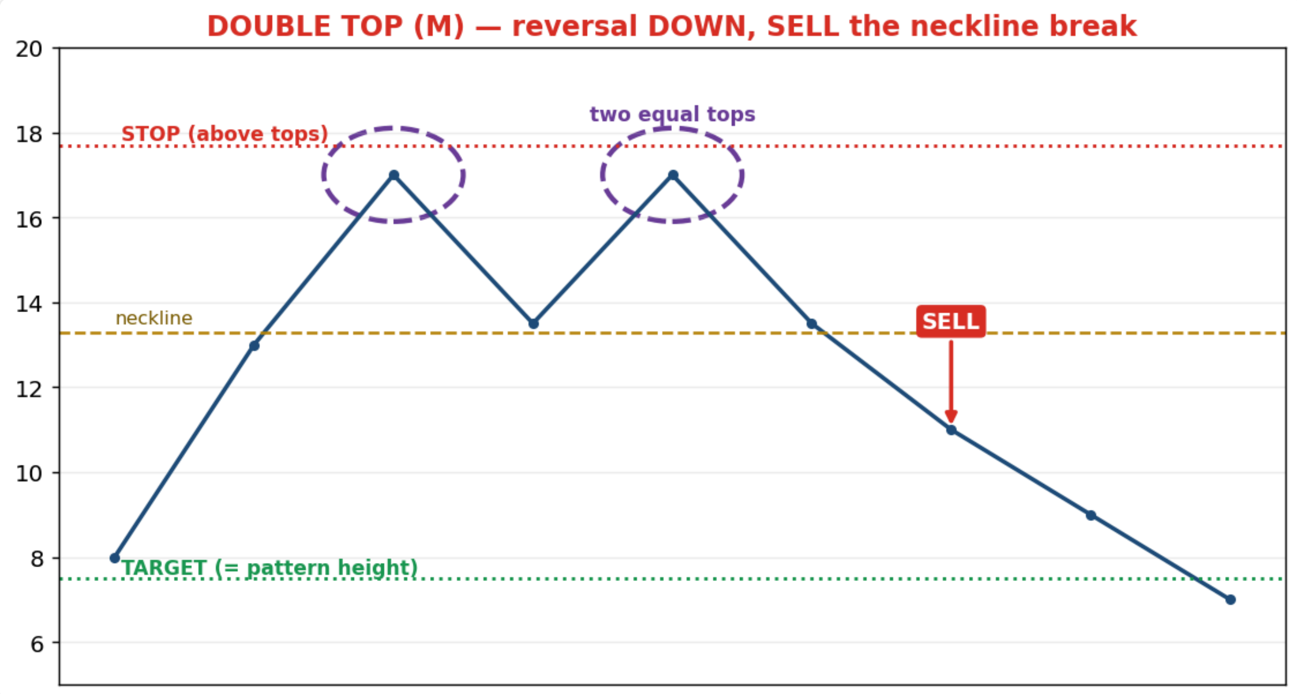
**The psychology:** Sellers attack the low twice and fail both times — they realise demand won't break, so selling dries up and the reversal begins. Sellers who shorted the second dip get trapped and must buy back.

**How to use / trade it:** BUY when price closes above the middle bounce high (the neckline); stop below the bottoms; target = the pattern's height added to the breakout point.

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** Buy when price closes above the middle bounce high (the neckline).
- **Stop-loss:** Just below the two bottoms.
- **Target / when to sell:** The pattern's height (bottoms-to-neckline) added to the breakout point.

## Double Top (M) — reversal down




**What it is:** Price rises to a level, pulls back, rises to the SAME level again, then falls — an 'M' shape.

**Why it works:** A ceiling that rejects price twice proves strong selling supply; the second failure makes buyers give up.

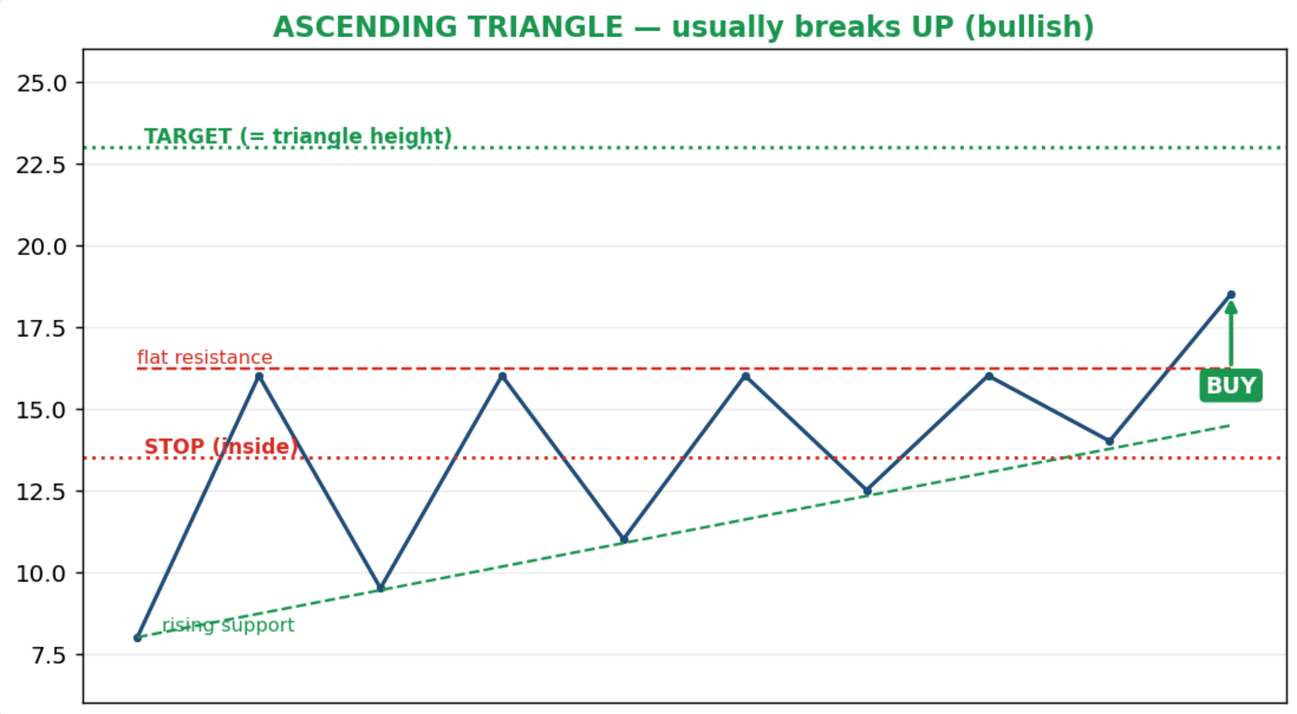
**The psychology:** Buyers push the high twice and fail — they realise supply won't break, so buying dries up and the decline begins. Buyers who chased the second push get trapped and bail.

**How to use / trade it:** SELL when price closes below the middle pullback low (the neckline); stop above the tops; target = the pattern's height below the breakdown.

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** Sell / exit longs when price closes below the middle low (the neckline).
- **Stop-loss:** Just above the two tops.
- **Target / when to sell:** The pattern's height projected below the breakdown.

## Ascending Triangle — usually breaks UP




**What it is:** A flat resistance line on top with rising lows underneath, squeezing into the corner.

**Why it works:** Buyers keep paying higher prices (rising lows) while sellers defend one ceiling — eventually buyers absorb all the supply and break out.

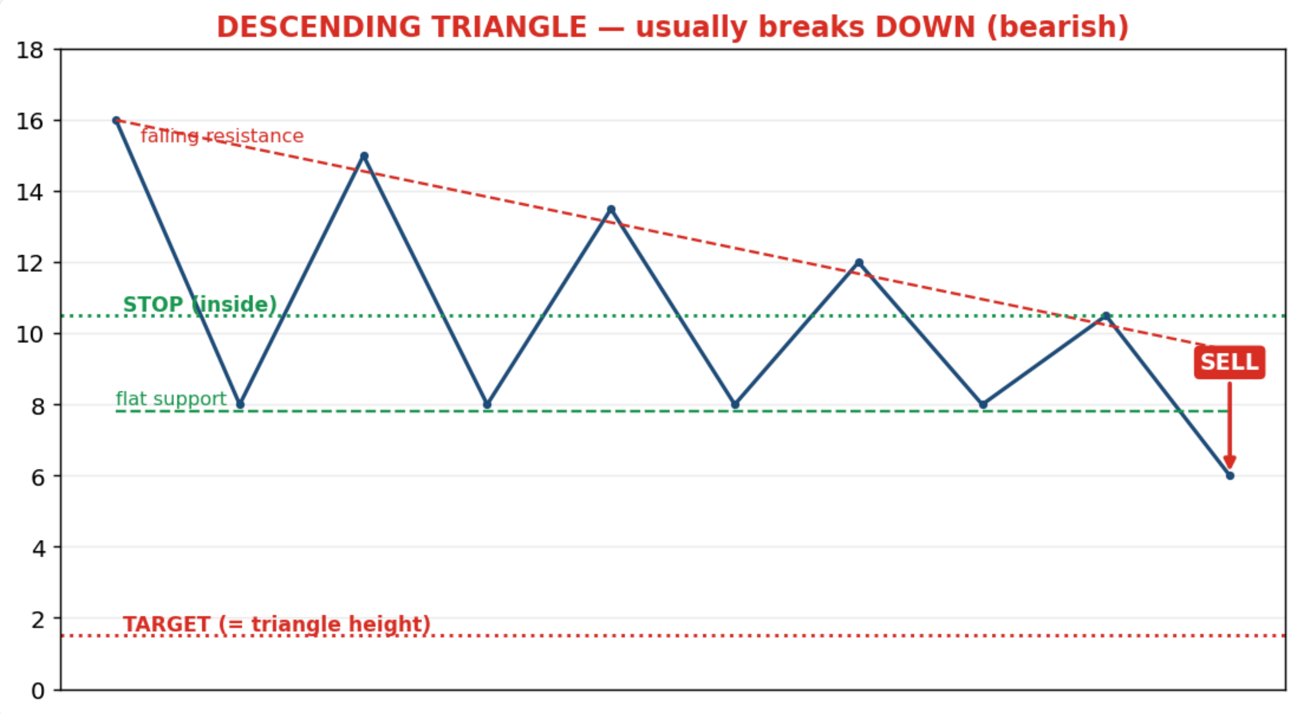
**The psychology:** The rising lows show growing, impatient demand; the flat top is a wall of sell orders. When that wall is finally eaten through, trapped sellers plus FOMO buyers fuel the upside break.

**How to use / trade it:** BUY the breakout above the flat top, ideally on a volume surge; stop just inside the triangle; target = the triangle's height added to the breakout.

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** Buy the close above the flat resistance top, ideally on a volume surge.
- **Stop-loss:** Just inside the triangle.
- **Target / when to sell:** The triangle's height added to the breakout point.

### Descending Triangle — usually breaks DOWN




**What it is:** A flat support line on the bottom with falling highs above, squeezing downward.

**Why it works:** Sellers keep accepting lower prices (falling highs) while buyers defend one floor — eventually sellers break it.

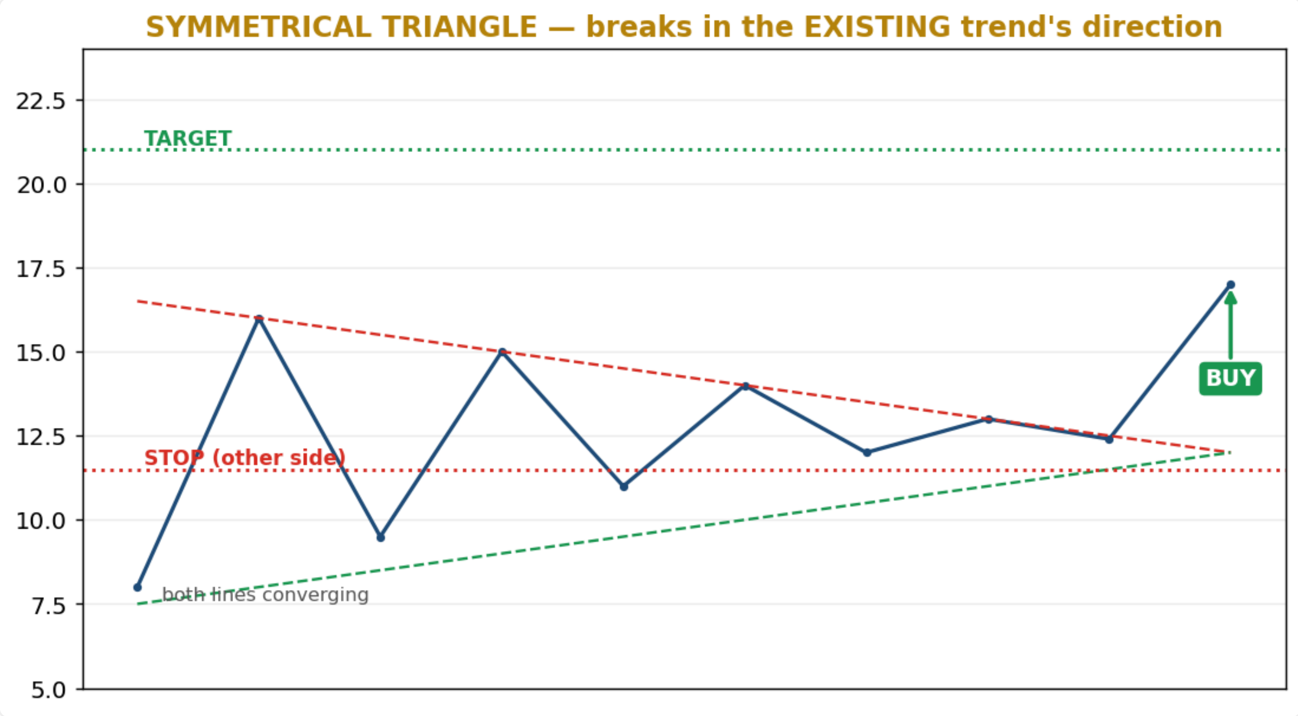
**The psychology:** Falling highs show weakening demand; the flat bottom is a shrinking wall of buy orders. When it breaks, trapped buyers bail and the decline accelerates.

**How to use / trade it:** SELL the breakdown below the flat bottom; stop just inside the triangle; target = the triangle's height below the breakdown.

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** Sell / short the close below the flat support bottom.
- **Stop-loss:** Just inside the triangle.
- **Target / when to sell:** The triangle's height projected below the breakdown.

## Symmetrical Triangle — breaks WITH the trend




**What it is:** Both lines converge — lower highs AND higher lows squeezing to a point.

**Why it works:** Buyers and sellers both compress; the breakout reveals whose conviction was real — usually the prior trend wins.

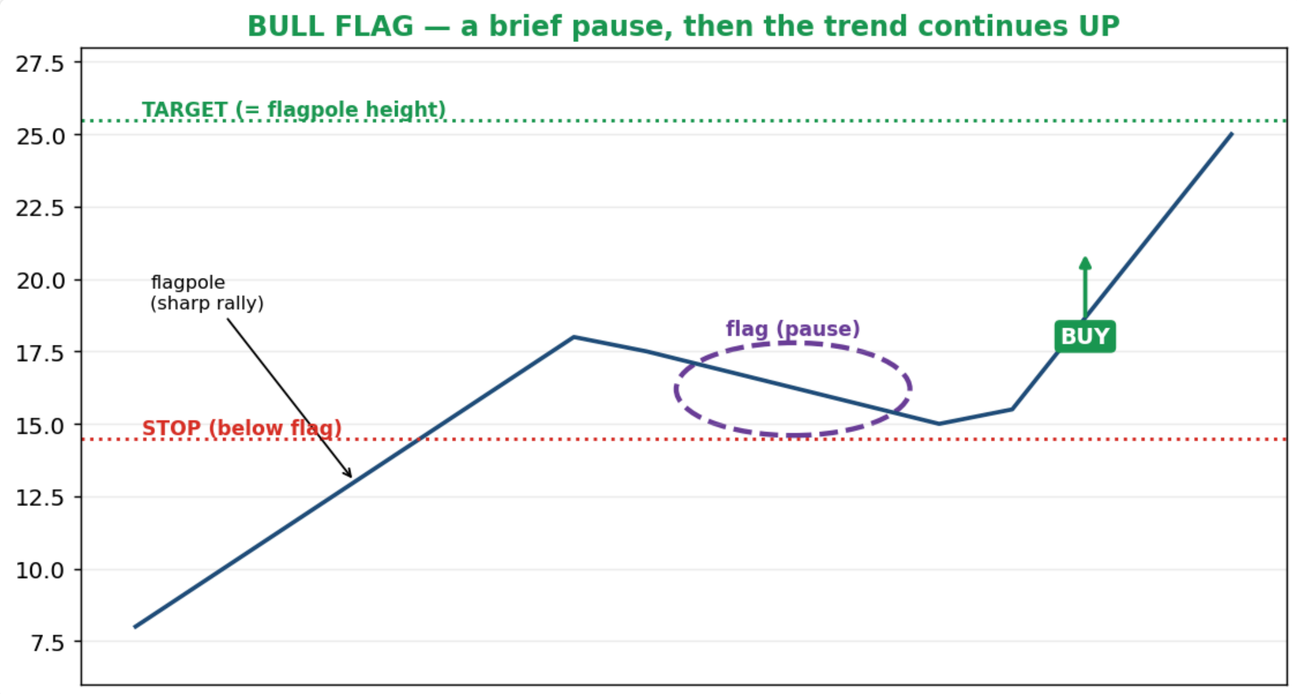
**The psychology:** It's a coiled spring of indecision; volatility contracts to a calm, then releases violently once one side finally gives way.

**How to use / trade it:** Trade the breakout in its direction (favour the prior trend), ideally with volume; stop on the opposite side; target = the triangle's widest height.

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** Trade the breakout in its direction (favour the prior trend), with volume.
- **Stop-loss:** On the opposite side of the triangle.
- **Target / when to sell:** The triangle's widest (left-hand) height.

## Bull Flag — continuation up




**What it is:** A sharp rally (the 'flagpole'), then a small sideways/down drift (the 'flag'), then another leg up.

**Why it works:** Strong moves pause to let early buyers take profit; once weak hands are out, the dominant buyers resume.

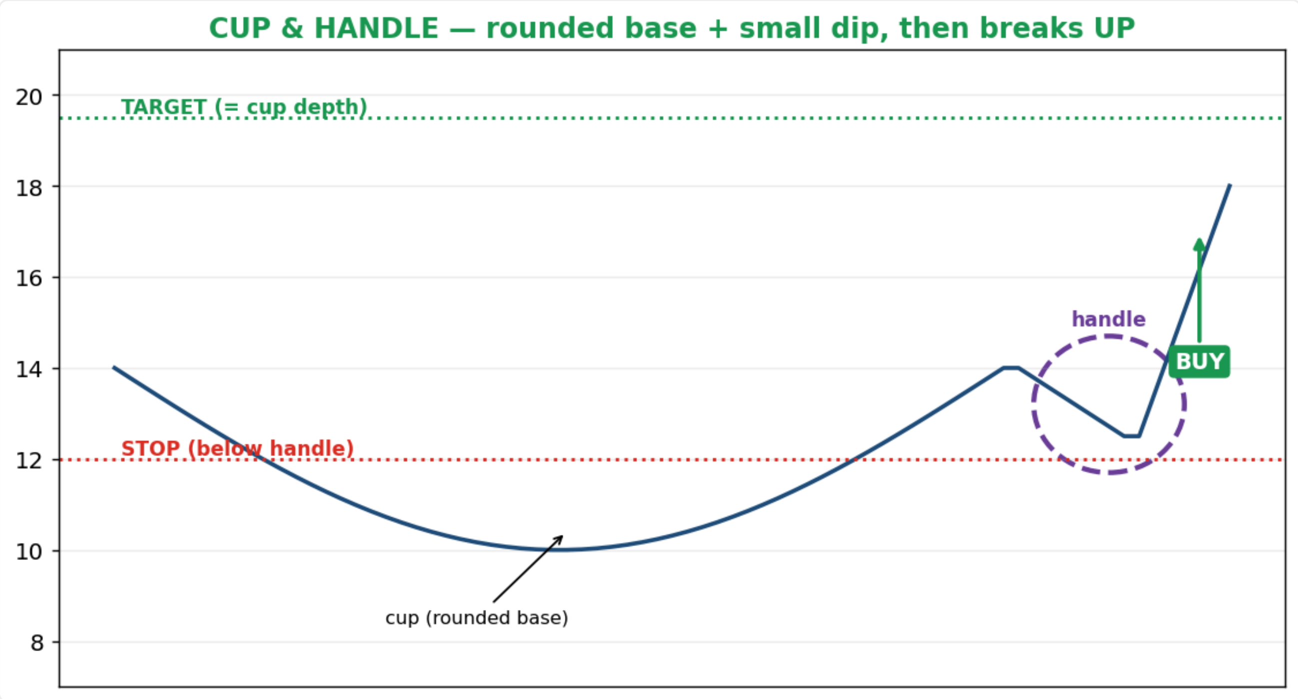
**The psychology:** The pullback shakes out nervous holders while the big players stay put — so the trend continues after the breather.

**How to use / trade it:** BUY the breakout above the flag; stop below the flag low; target = the flagpole height added to the breakout point.

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** Buy the breakout above the flag.
- **Stop-loss:** Below the flag low.
- **Target / when to sell:** The flagpole height added to the breakout point.

## Cup & Handle — continuation up




**What it is:** A rounded 'U' base (the cup), then a small dip (the handle), then a breakout upward.

**Why it works:** The rounded bottom shows sentiment slowly shifting from selling to buying; the handle is a final shakeout.

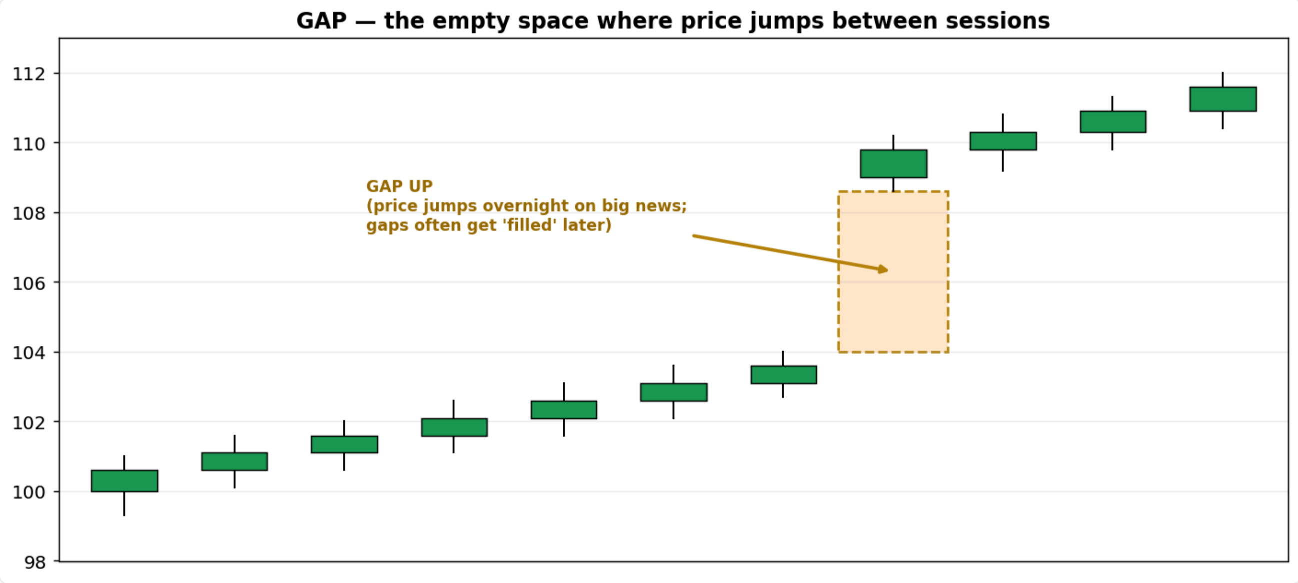
**The psychology:** Despair fades to hope across the cup; the small handle dip flushes the last weak holders right before the breakout.

**How to use / trade it:** BUY the breakout above the handle's high; stop below the handle; target = the cup's depth added to the breakout point.

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** Buy the breakout above the handle's high.
- **Stop-loss:** Below the handle.
- **Target / when to sell:** The cup's depth added to the breakout point.

## Gaps




**What it is:** Empty space on the chart where price opens far from the previous close — usually on overnight news.
**Why it works:** A gap marks a sudden supply/demand imbalance strong enough that no trading happened in between.
**The psychology:** Gaps are pure emotion crystallised: a breakaway gap = new information genuinely repricing the asset; an exhaustion gap = the last euphoric buyers or panicked sellers piling in at the end of a move. Many gaps 'fill' later because the initial reaction was an emotional overshoot that rational traders fade.
**How to use / trade it:** Breakaway gaps (with volume) often start strong trends — trade with them; exhaustion gaps often mark ends — fade them. Watch for the gap to fill as a target.

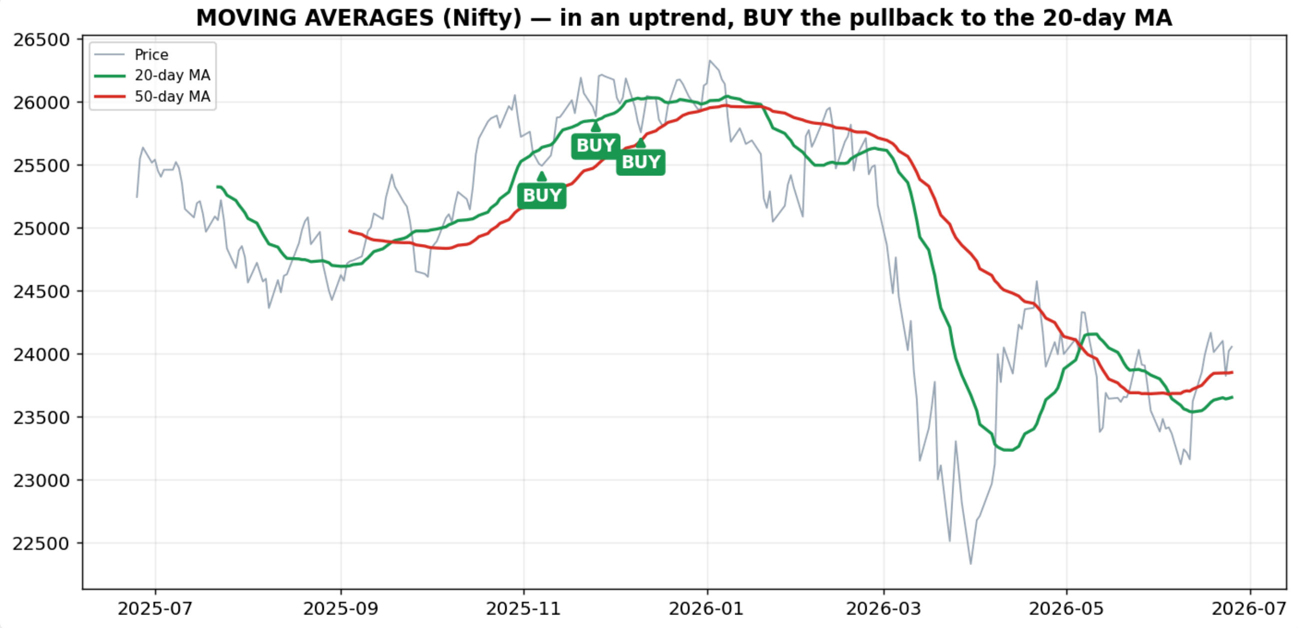
**Every part, explained simply:** A gap is a blank space where price jumps overnight (opens far from the prior close), usually on big news. A **breakaway gap** with heavy volume often starts a strong new trend (trade with it). An **exhaustion gap** at the end of a long move (last euphoric/panicked traders) often reverses. Gaps frequently get 'filled' as price drifts back to where the jump began.

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** Breakaway gap (with volume): enter in the gap's direction on the open or first small pullback. Exhaustion gap: fade it — trade toward the gap getting filled.
- **Stop-loss:** On the far side of the gap (a move back through it kills the idea).
- **Target / when to sell:** Trend continuation for a breakaway gap; the prior close (gap fill) for an exhaustion gap.

## PART 3 — Indicators (and why each works)

### Moving averages

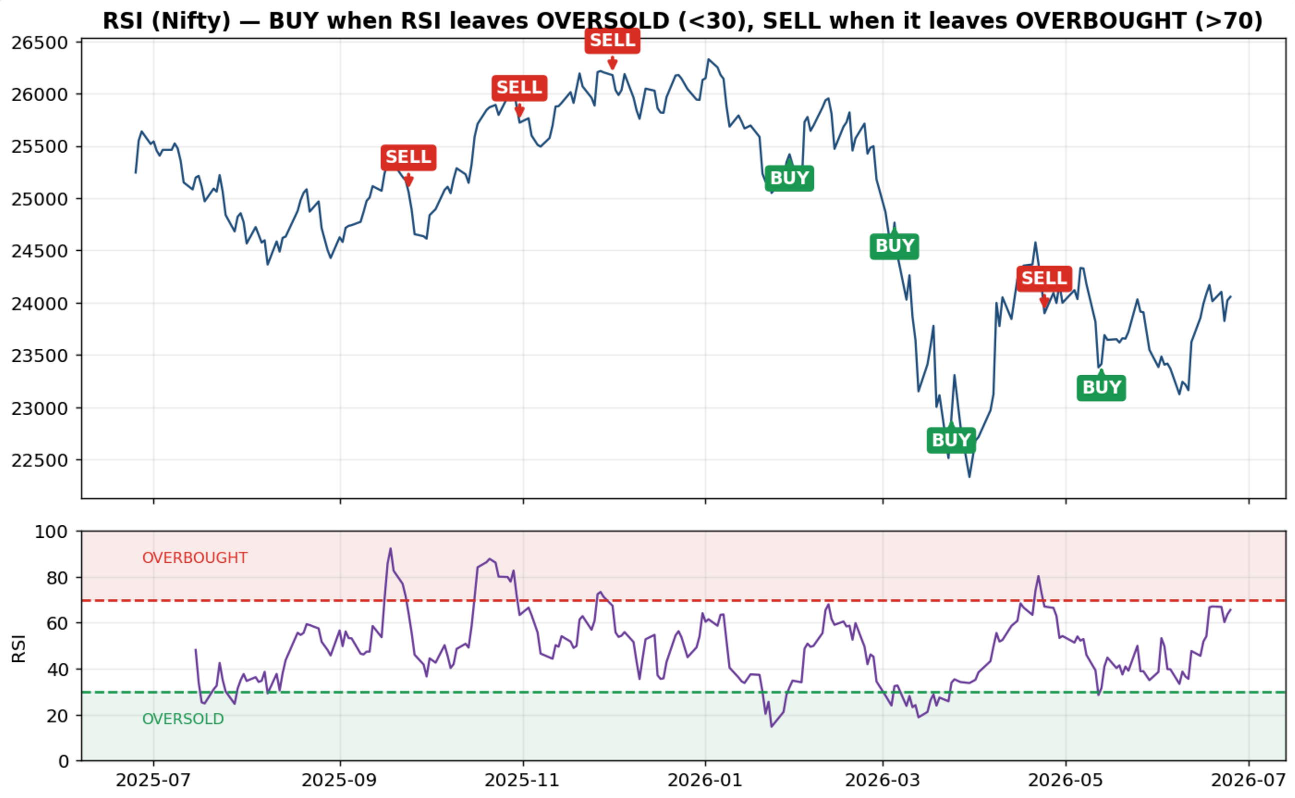



**What it is:** The average closing price over the last N days, re-plotted daily, smoothing out noise.
**Why it works:** It represents the average price participants paid recently — a moving 'consensus value' that the market gravitates around.
**The psychology:** When price is above the MA, the average holder is in profit and feels confident (bullish bias); below it, holders are underwater and nervous. Traders treat the MA as a fair-value reference and buy dips to it, which is why it acts as dynamic support. A golden cross (fast above slow) means the recent average cost has risen above the long-term — sentiment has structurally turned up.
**How to use / trade it:** Use price-vs-MA for trend bias and MA crossovers for shifts; the 20/50/200-day are the most-watched (self-fulfilling because everyone uses them).

> **Every part, explained simply:** A moving average is the average of the last N closing prices, recalculated each day so it 'moves'. A **fast** MA (e.g. 20-day) hugs price and reacts quickly; a **slow** MA (50- or 200-day) shows the big trend. **SMA** weights every day equally; **EMA** weights recent days more so it turns faster.
> **How it's calculated (formula):** `SMA = (P1 + P2 + ... + PN) / N`. EMA: `EMA = Price×k + EMA_prev×(1−k)`, where `k = 2/(N+1)`.
> **Worked example:** Last 5 closes 10, 12, 14, 16, 18 → SMA(5) = 70 ÷ 5 = **14**. For a 10-day EMA, k = 2/11 = 0.18; if yesterday's EMA was 100 and today's price 110 → 110×0.18 + 100×0.82 = **101.8**.

> **Trade plan — where to buy, stop and sell:**
> - **Buy / entry:** In an uptrend, buy a pullback to the 20- or 50-day MA when a bullish candle forms on it. A golden cross (50 above 200) is a positional buy signal.
> - **Stop-loss:** Below the MA / the pullback low.
> - **Target / when to sell:** The prior high; then trail with the MA and EXIT when price closes decisively below it.

## RSI — overbought / oversold



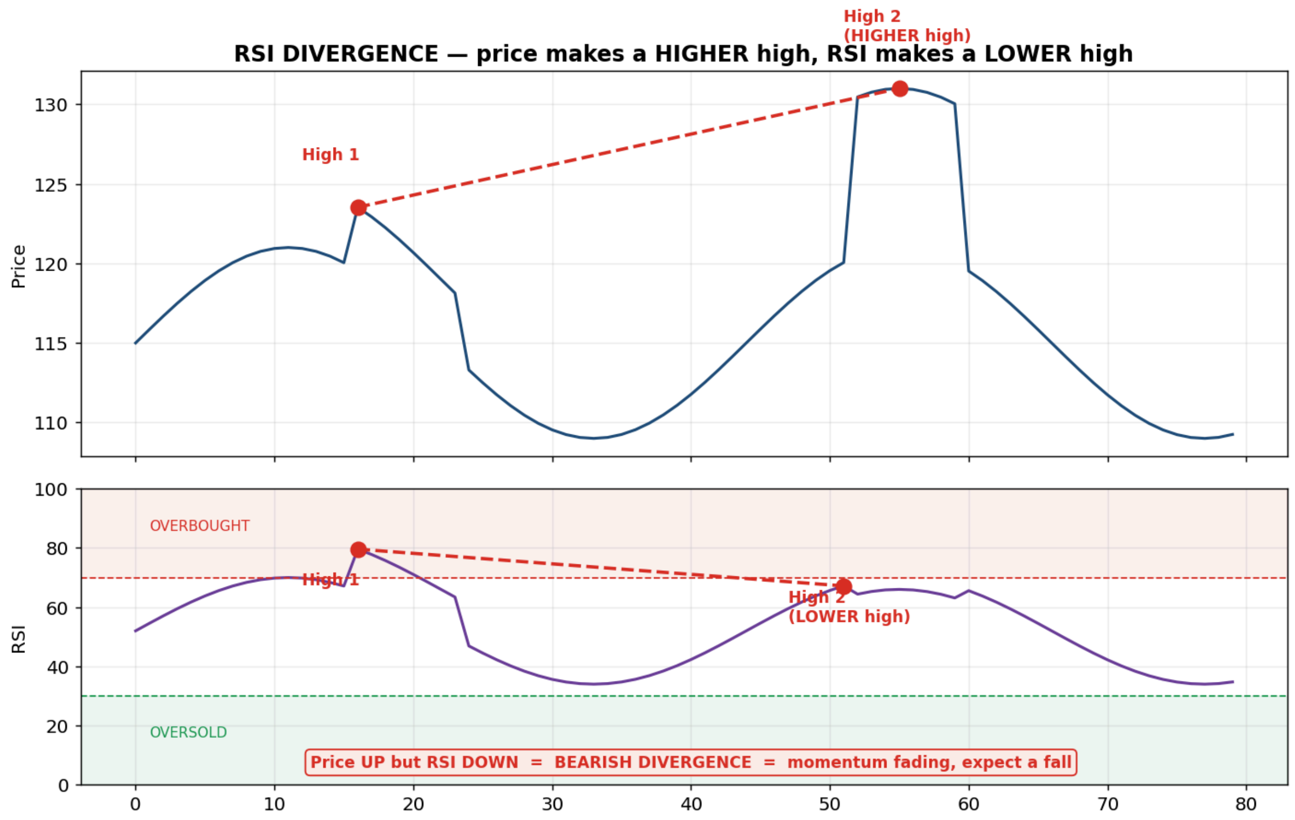

**What it is:** A 0–100 oscillator measuring the strength of recent gains versus losses (default 14 periods).
**Why it works:** It quantifies momentum — how one-sided the buying or selling has been — so you can spot when a move is overstretched.
**The psychology:** Above 70 (overbought) means buying has been so aggressive that nearly everyone who wanted in is already in — there's little fuel left, so a pause/pullback is likely. Below 30 (oversold) means panic selling has likely exhausted itself. **Divergence** is the deeper signal: if price makes a new high but RSI doesn't, the rally is being driven by fewer and weaker buyers — momentum is dying.
**How to use / trade it:** Use it as a warning light, not a trigger; in strong trends RSI can stay extreme for a long time, so combine it with price action.
**Interview depth:** Emphasise that 'overbought' ≠ 'sell now' — in a powerful uptrend shorting overbought RSI is a classic beginner mistake; divergence is the higher-quality signal.

> **Every part, explained simply:** RSI is one line travelling 0–100 in its own panel. It measures momentum: how big the up-days have been versus the down-days over the last 14 days. **Above 70** = overbought (risen too fast), **below 30** = oversold (fallen too fast), **50** = balance between buyers and sellers.
> **How it's calculated (formula):** `RS = Average Gain ÷ Average Loss` (over 14 days), then `RSI = 100 − 100/(1 + RS)`.
> **Worked example:** If the average gain is 1.0 and average loss 0.5 → RS = 2 → RSI = 100 − 100/3 = **66.7**. Sanity check: all up-days → RSI 100; all down-days → RSI 0; equal → 50.

> **Trade plan — where to buy, stop and sell:**
> - **Buy / entry:** In a range, buy when RSI climbs back above 30 (out of oversold). Strongest when paired with bullish divergence AND a break of short-term price structure.
> - **Stop-loss:** Below the recent swing low (or the divergence low).

• **Target / when to sell:** Sell when RSI reaches ~70 (overbought) in a range, or at the next resistance. Do NOT blindly short RSI > 70 in a strong uptrend.

## RSI Divergence — the early reversal warning




**What it is:** When price and the RSI line disagree — price makes a new high but RSI makes a lower high (or price makes a new low but RSI makes a higher low).

**Why it works:** RSI is built from momentum, so it can reveal that a new price high was made on weaker buying — the move is running on fumes before price actually turns.

**The psychology:** The crowd sees a fresh high and feels safe; the divergence quietly shows momentum fading underneath — smart money is often selling into that strength.

**How to use / trade it:** Confirm with a price break (a close below a recent swing low) before acting — divergence warns, price triggers.

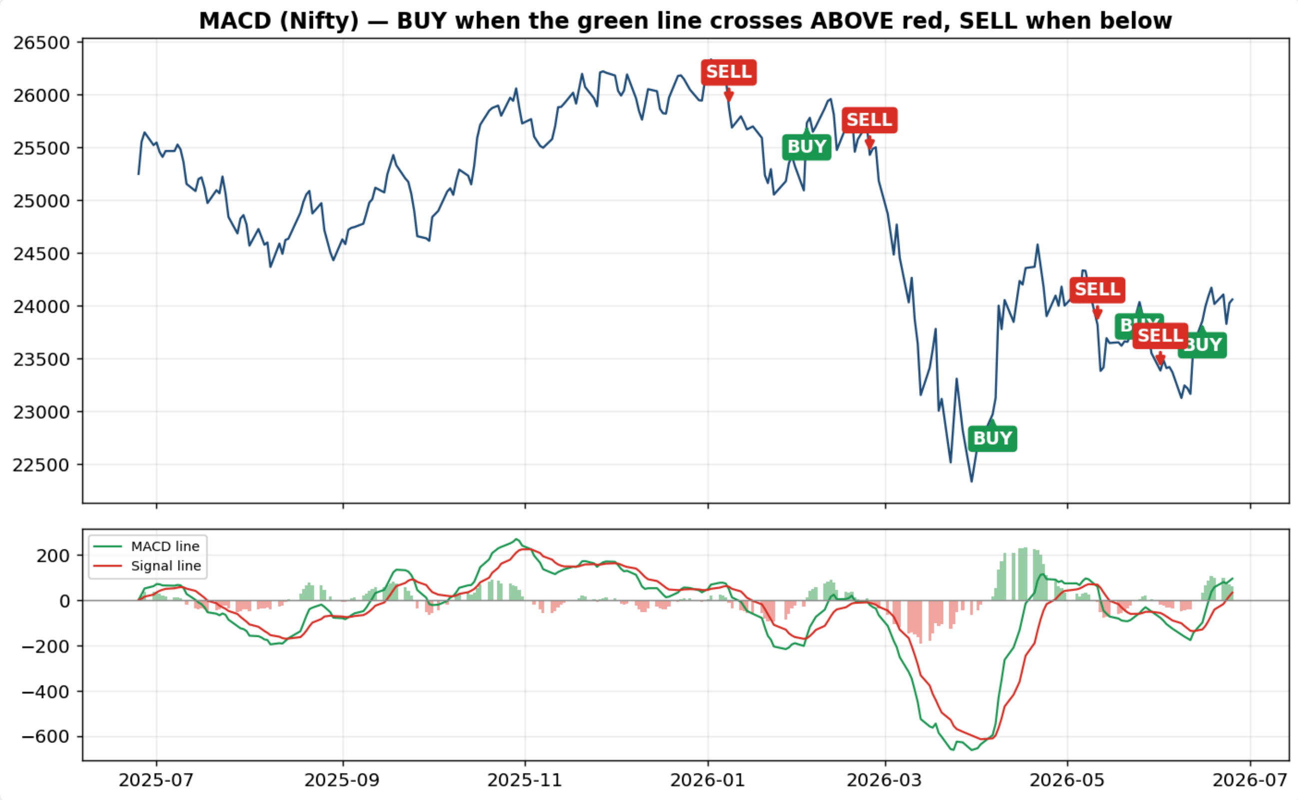
**Interview depth:** Regular divergence warns of a reversal; hidden divergence signals trend continuation.

**Every part, explained simply:** Divergence = price and the RSI line disagree. To spot it, follow the red dots on the chart:

- **Step 1:** mark the last two major **highs** on the PRICE chart (top).
- **Step 2:** mark the matching two highs on the RSI line (bottom).
- **Step 3:** compare them —
  – Price **Higher High** + RSI **Lower High** → **Bearish divergence** (rally weakening → expect a fall).
  – Price **Lower Low** + RSI **Higher Low** → **Bullish divergence** (fall weakening → expect a bounce).

Why it matters: price made a new high, but RSI shows the momentum behind it was actually weaker — fewer, tired buyers. An early warning the trend may reverse.

## MACD — trend + momentum




**What it is:** The gap between a fast (12) and slow (26) EMA, with a 9-EMA signal line and a histogram of the gap.
**Why it works:** It captures the rate at which the short-term average is pulling away from the long-term — i.e., accelerating or fading momentum.
**The psychology:** A bullish crossover means short-term buying pressure has overtaken the longer-term trend — the balance of power is shifting to buyers in real time. The histogram shrinking even while price rises warns that momentum is fading before price actually turns.
**How to use / trade it:** Use crossovers for entries/exits and histogram + divergence for early warnings; best as confirmation alongside trend and levels.

**Every part, explained simply:** MACD has THREE parts:

- **MACD line** (main line) = a fast 12-day EMA minus a slow 26-day EMA — it measures how far the short-term average has pulled away from the long-term one.
- **Signal line** = a 9-day EMA OF the MACD line — a smoothed, slower copy used as the trigger.
- **Histogram** (the bars) = the MACD line minus the Signal line. Tall bars = lines far apart = strong momentum; shrinking bars = lines converging = momentum fading. Bars flip above zero exactly when the MACD line crosses above the signal line.

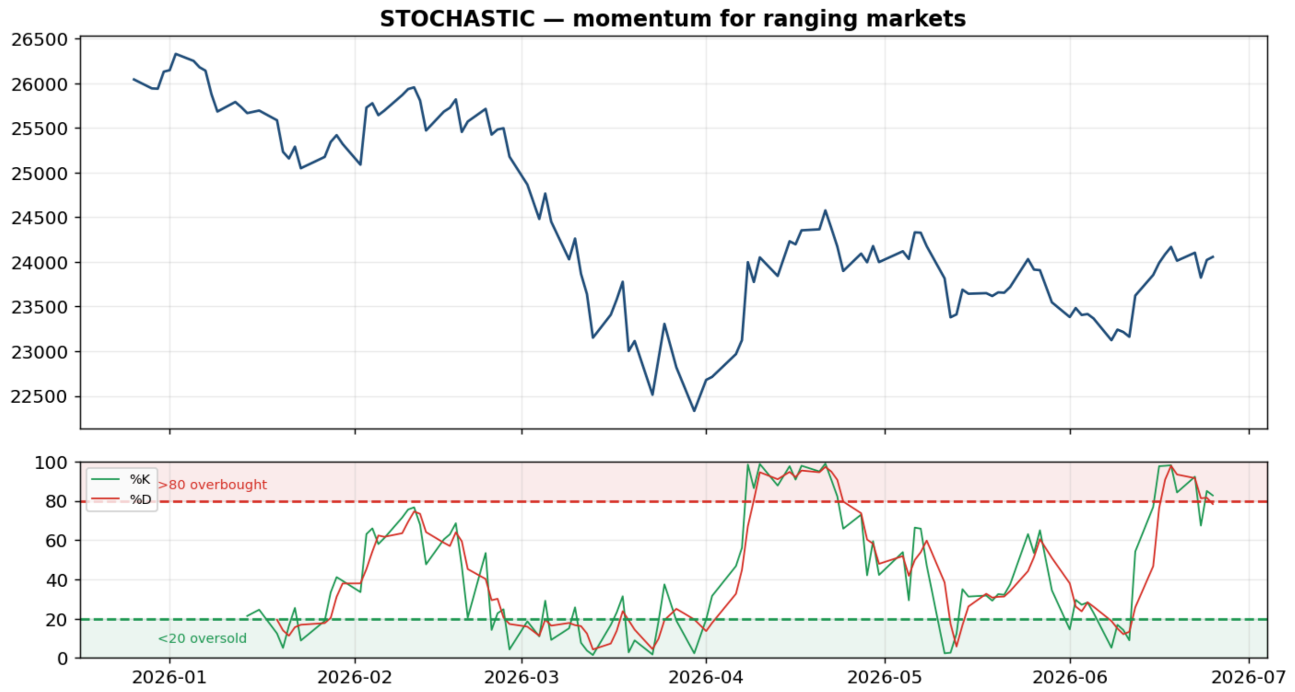
**How it's calculated (formula):** `MACD = EMA(12) − EMA(26)`; `Signal = EMA(9) of MACD`; `Histogram = MACD − Signal`.
**Worked example:** If the 12-EMA = 1020 and 26-EMA = 1000 → MACD = +20. If the signal line = 15 → histogram = +5 (positive = bullish; the MACD line is above its signal).

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** Go long on a bullish crossover (MACD line crosses above the signal line), ideally above the zero line.
- **Stop-loss:** Below the recent swing low.

• **Target / when to sell:** Sell/exit on the bearish crossover (MACD back below signal), or at the next resistance. A shrinking histogram = tighten your stop.

## Stochastic oscillator



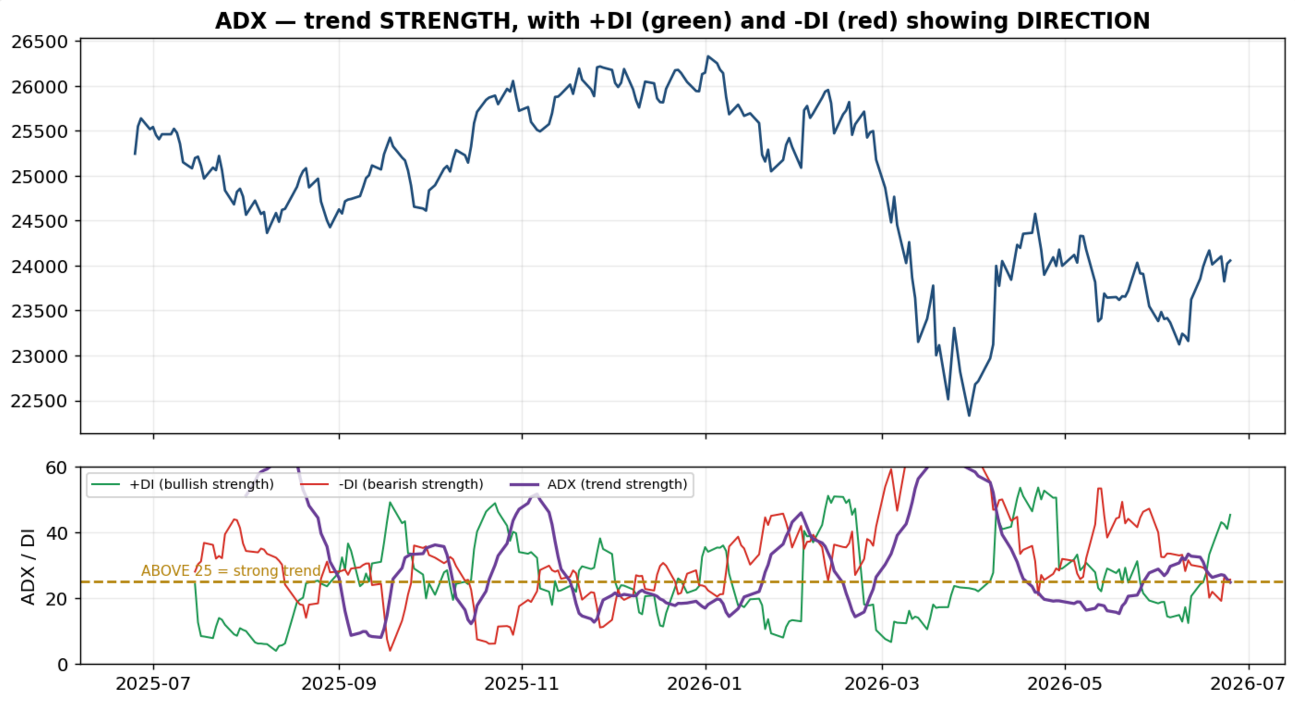

**What it is:** Shows where the close sits within the recent high-low range (0–100), with %K and %D lines.
**Why it works:** Closing near the top of the range signals strength; near the bottom, weakness — a quick read on momentum, especially in ranges.
**The psychology:** In an uptrend price tends to close near its highs (confident buyers); when it starts closing in the lower part of the range despite higher prices, buyers are tiring. Overbought/oversold zones flag emotional extremes in range-bound markets.
**How to use / trade it:** Best in sideways markets; use %K/%D crossovers in the 80/20 zones, and watch for divergence.

**Every part, explained simply:** Two lines, 0–100: **%K** answers 'where did price close inside its recent range?' (0 = at the very low, 100 = at the very high). **%D** is a 3-day average of %K — the smoother trigger line. Above 80 = closing near the highs (overbought); below 20 = near the lows (oversold).
**How it's calculated (formula):** `%K = (Close − Lowest Low) ÷ (Highest High − Lowest Low) × 100` (over 14 days); `%D = 3-day average of %K`.
**Worked example:** If over 14 days the low was 40, the high 60 and today's close 58 → %K = (58−40)/(60−40)×100 = **90** (closed near the top = strong).

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** In a range, buy when %K crosses above %D while below the 20 line (oversold).
- **Stop-loss:** Below the recent swing low.
- **Target / when to sell:** Sell when stochastic crosses down from above the 80 line (overbought).

## ADX — trend STRENGTH




**What it is:** A 0–100 line measuring how strong a trend is, regardless of direction.

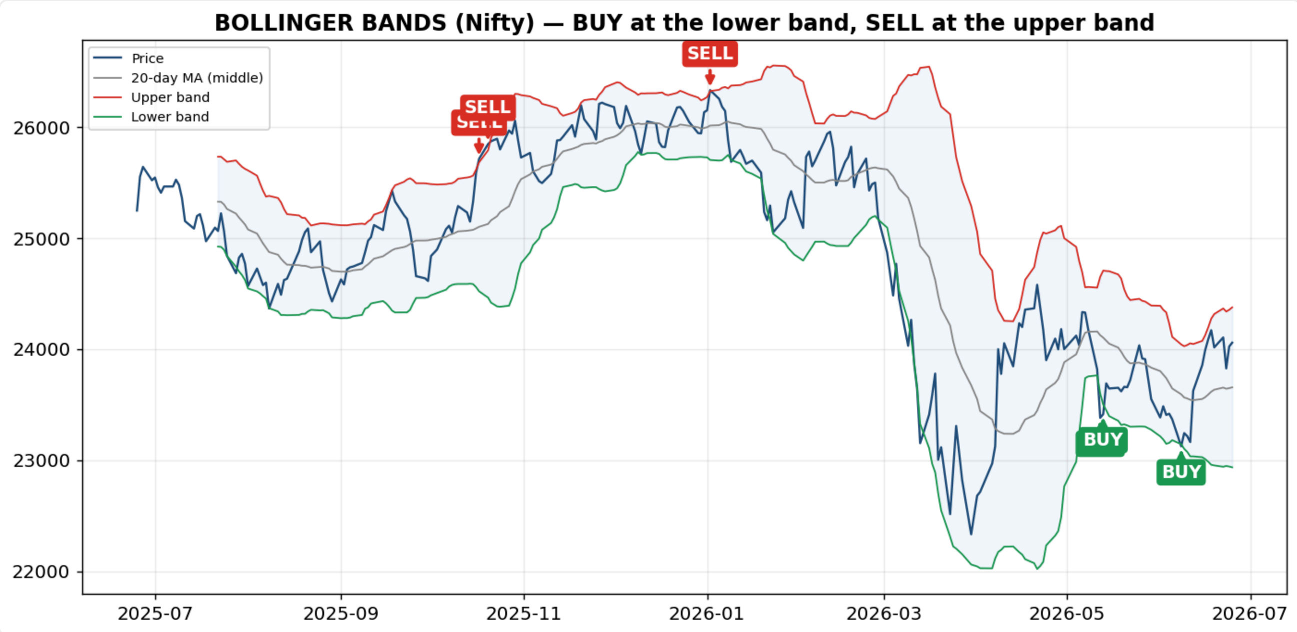
**Why it works:** Many tools only work in the right regime — ADX tells you whether to use trend tools or range tools.

**The psychology:** A high ADX means participants broadly agree on direction (strong conviction, persistent trend); a low ADX means disagreement/indecision (choppy range). It's a measure of crowd consensus strength.

**How to use / trade it:** Above 25 → trade with the trend (let winners run); below 20 → switch to range tactics (fade extremes). Don't use RSI overbought-as-sell when ADX is high — the trend will keep pushing.

> **Every part, explained simply:** ADX comes with TWO companion lines that show DIRECTION:
>
> - **+DI line (green)** = the strength of UPWARD (bullish) price movement.
> - **−DI line (red)** = the strength of DOWNWARD (bearish) price movement.
>
> When +DI (green) is above −DI (red), buyers are in control; when −DI is on top, sellers are. The **ADX line** itself (0–100) measures only how STRONG the trend is, not its direction: above 25 = strong trend, below 20 = weak/sideways.
>
> **How it's calculated (formula):** `+DI = 100 × smoothed(+DM) ÷ ATR`; `−DI = 100 × smoothed(−DM) ÷ ATR`; `DX = 100 × |+DI − −DI| ÷ (+DI + −DI)`; `ADX = smoothed average of DX`.
>
> **Worked example:** If +DI = 30 and −DI = 10 → DX = 100 × |30−10| ÷ (30+10) = 100 × 20/40 = **50** — a strong uptrend (buyers clearly dominating).

## Bollinger Bands — volatility




**What it is:** A 20-day average with bands set 2 standard deviations above and below it.
**Why it works:** Two standard deviations statistically contain ~95% of recent prices, so touches of the bands flag stretched conditions, and band WIDTH measures volatility.
**The psychology:** A 'squeeze' (very narrow bands) reflects low disagreement and complacency — a coiled spring; when new information arrives, the pent-up move releases violently (expansion). Price hugging the upper band shows persistent strength, not just 'overbought'.
**How to use / trade it:** Trade squeeze breakouts in the breakout direction; in a range, fade band touches back to the middle average.

**Every part, explained simply:** Developed by John Bollinger in the 1980s to measure volatility and flag overbought/oversold conditions. THREE lines wrap around price:

- **Middle band** = a 20-day Simple Moving Average — the baseline for the intermediate trend.
- **Upper band** = middle + 2 standard deviations — touching it signals potential **overbought** territory.
- **Lower band** = middle − 2 standard deviations — touching it signals potential **oversold** territory.

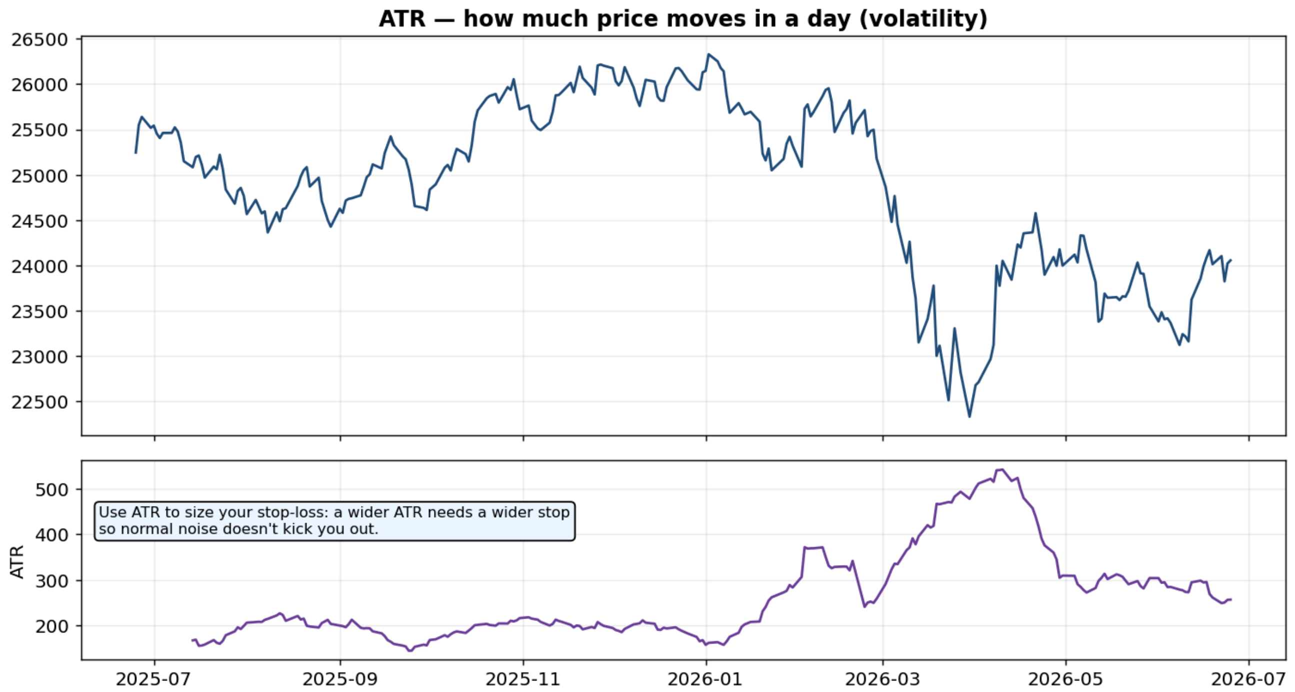
**Standard deviation (σ)** measures how far prices stray from their average — i.e. volatility.
**Three core ideas:** (1) **Volatility — squeeze & bulge:** the bands widen when volatility is high and tighten when it's low; a tight **squeeze** warns of an explosive breakout coming. (2) **Mean reversion — the bounce:** ~95% of price action stays inside the bands, so a tag of the upper/lower band often snaps back toward the middle. (3) **Trend confirmation — walking the band:** in a strong trend price can 'ride' the upper (or lower) band instead of reversing — so don't blindly fade a band touch in a powerful trend. Best used WITH a momentum tool like RSI or MACD to avoid false signals.
**How it's calculated (formula):** `Middle = SMA(20)` ; `Upper = Middle + 2σ` ; `Lower = Middle − 2σ` (σ = standard deviation of the last 20 closes).
**Worked example:** If the 20-day average = 100 and σ = 5 → Upper = 100 + 10 = **110**, Lower = 100 − 10 = **90**. About 95% of prices stay between the bands.

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** Range play: buy at the LOWER band when a bullish candle prints. Squeeze play: buy on a close OUTSIDE the upper band after the bands have pinched tight.
- **Stop-loss:** Range: below the lower band. Breakout: back at the middle band (20-MA).

• **Target / when to sell:** Range: the middle or upper band. Breakout: ride until price closes back inside the bands.

## ATR — how much it moves

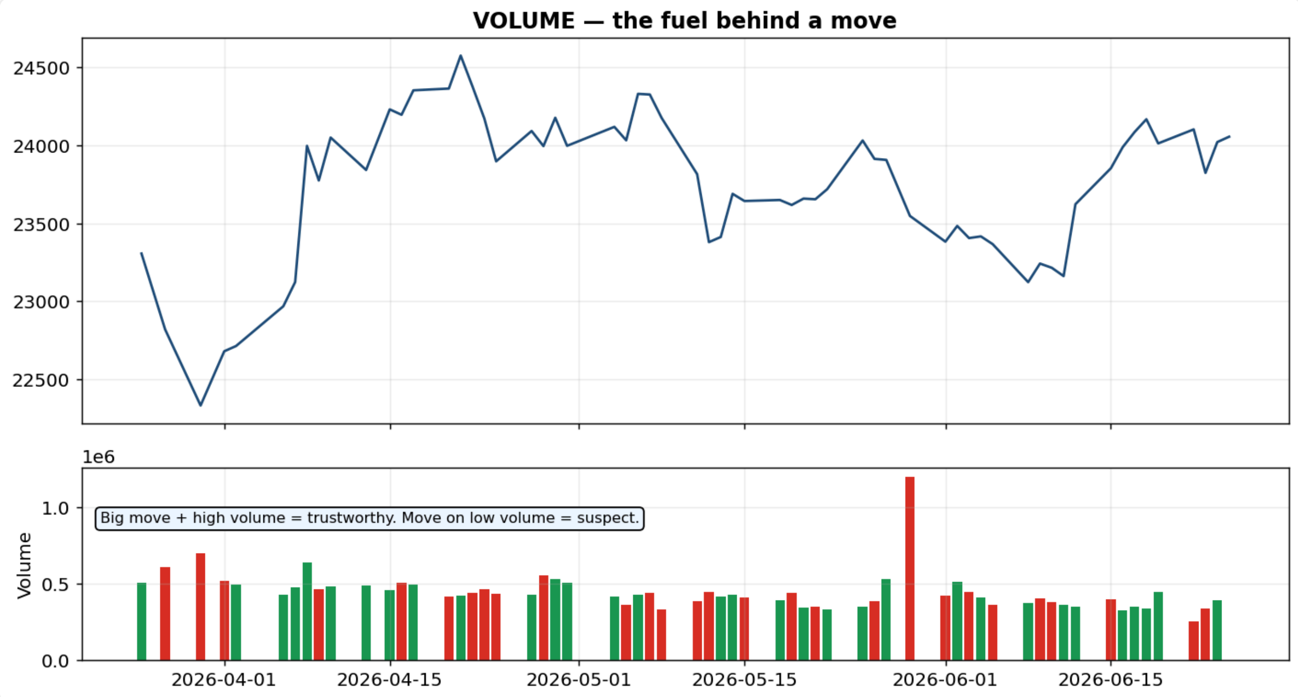



**What it is:** Average True Range — the typical size of a day's move, including gaps (pure volatility, no direction).
**Why it works:** Risk must be scaled to volatility; ATR gives you that yardstick objectively.
**The psychology:** Volatility is emotional intensity — ATR rises when fear/excitement spikes. A stop placed without regard to ATR will be too tight in a volatile name (stopped out by noise) or too loose in a calm one.
**How to use / trade it:** Set stops at ~1.5–2× ATR from entry and size positions so each trade risks a similar rupee amount regardless of the asset's volatility.

**Every part, explained simply:** ATR tells you how much an asset typically moves in ONE day, in points/rupees — a pure volatility number with no direction. It uses the 'True Range', the biggest of three measures so it also captures overnight gaps.
**How it's calculated (formula):** `True Range = the largest of: (High − Low), |High − PrevClose|, |Low − PrevClose|` ; `ATR = 14-day average of True Range`.
**Worked example:** Today's High = 105, Low = 100, yesterday's Close = 102 → TR = max(5, |105−102|=3, |100−102|=2) = **5**. Average 14 of these for the ATR.

**Trade plan — where to buy, stop and sell:**
- **Buy / entry:** ATR is not an entry trigger — it SIZES your risk on whatever setup you take.
- **Stop-loss:** Place the stop about 1.5–2× ATR away from entry, so normal daily noise won't knock you out.
- **Target / when to sell:** Set the target in ATR multiples too (e.g. 3× ATR) so reward-to-risk stays 2:1 or better.

## Volume — the fuel




**What it is:** The number of shares/contracts traded; shown as bars under price.

**Why it works:** Price tells you the direction; volume tells you the conviction behind it.

**The psychology:** A big move on heavy volume means many participants agree — real money is committed, so it's likely to continue. The same move on thin volume means few believe it; it's fragile and often reverses. Rising volume into a breakout confirms genuine demand.

**How to use / trade it:** Demand volume confirmation on breakouts; be sceptical of moves and patterns that complete on weak volume.

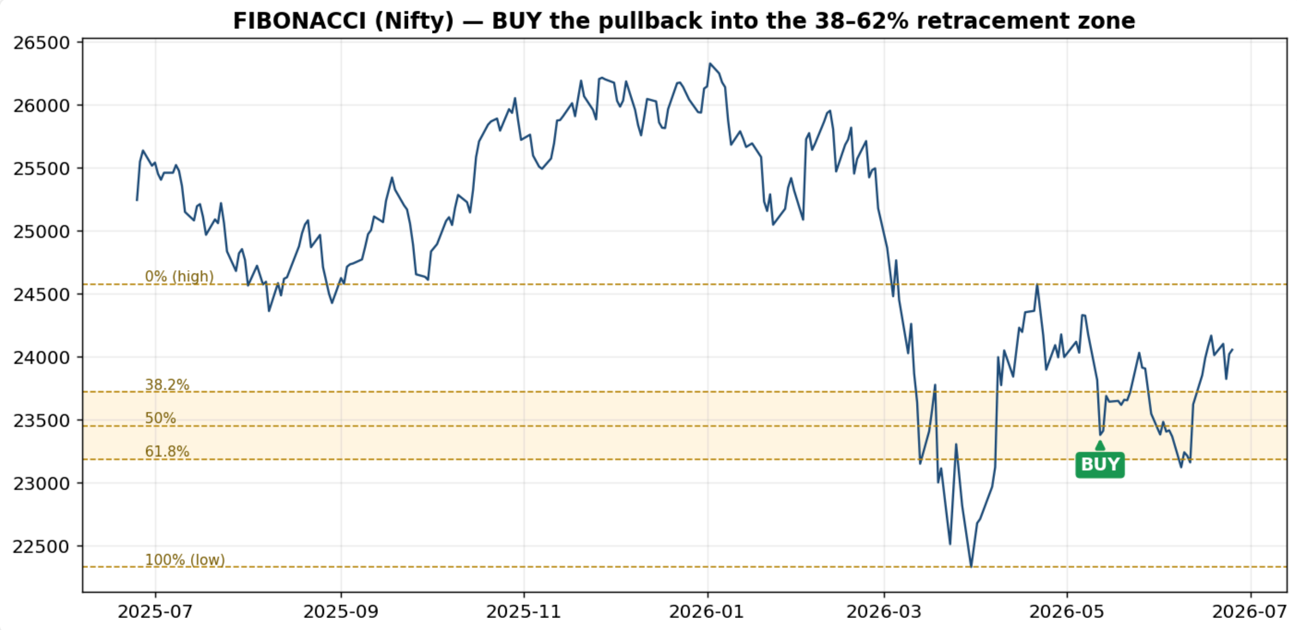
**Every part, explained simply:** Volume is simply the number of shares/contracts traded each day, drawn as bars under price. No formula — you compare bar heights. A tall bar = many participants (high conviction); a short bar = few (weak conviction).

**Worked example:** A breakout candle with a volume bar 2–3× the recent average = trustworthy. The same breakout on a below-average bar = likely a fake-out.

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** Only trust a breakout entry when volume EXPANDS on the breakout candle.
- **Stop-loss:** Standard (below the broken level).
- **Target / when to sell:** Standard target — but exit early if the move is running on fading volume (no conviction).

## Fibonacci retracement



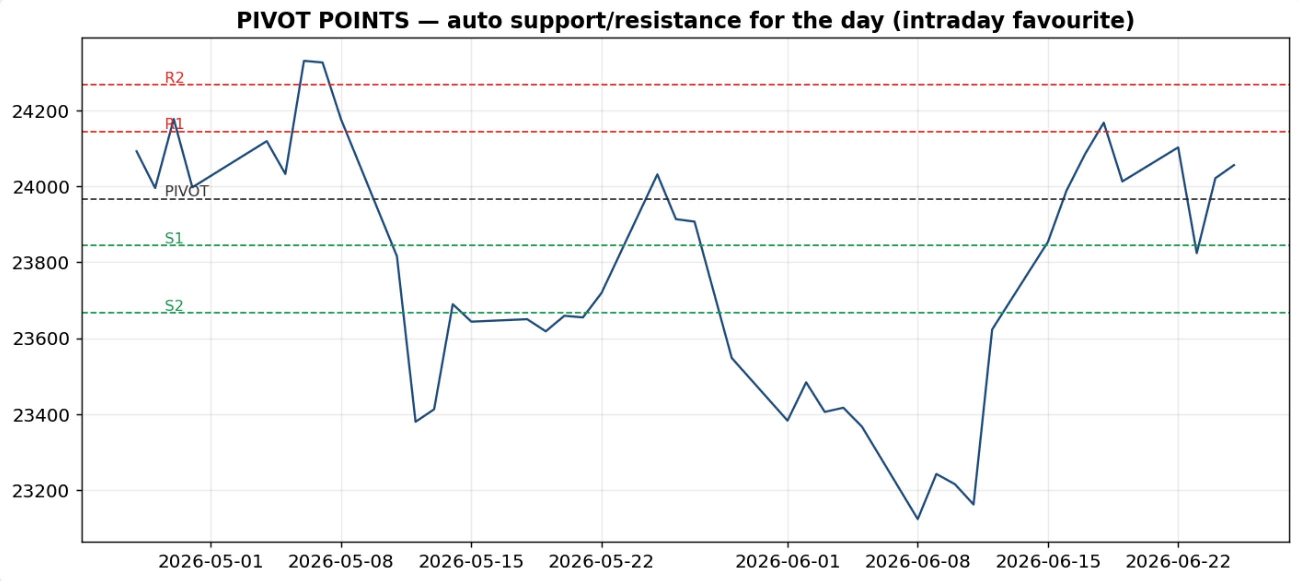

**What it is:** Horizontal levels at 23.6%, 38.2%, 50% and 61.8% of a prior move, marking likely pullback zones.
**Why it works:** It works through a mix of natural profit-taking rhythm and self-fulfilment — so many traders watch these levels that orders cluster there.
**The psychology:** After a rally, some buyers take profit, causing a dip. The 38.2–61.8% zone is where the pullback has given back enough to shake out weak hands and tempt new buyers who missed the move — so demand re-emerges there. A 61.8% retracement is the deepest 'healthy' pullback before the trend is in doubt.
**How to use / trade it:** In an uptrend, buy dips into the 38–62% zone, especially where a Fib level lines up with support or a moving average (confluence = higher probability).

**Every part, explained simply:** You draw the Fib tool from a recent swing LOW to a recent swing HIGH and it auto-plots horizontal lines at 23.6%, 38.2%, 50% and 61.8% of that move — the levels where a pullback tends to pause. The numbers come from the Fibonacci sequence (1,1,2,3,5,8,13…); dividing neighbours gives ~61.8%.
**How it's calculated (formula):** `Level = High − (percentage × (High − Low))`.
**Worked example:** Move from Low 100 to High 200 (range 100): 38.2% level = 200 − 0.382×100 = **161.8**; 61.8% level = 200 − 0.618×100 = **138.2**. Buyers often re-enter in the 138–162 zone.

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** In an uptrend, place buy orders in the 38.2%–61.8% retracement zone; enter on a bullish candle there (strongest where a Fib level overlaps support or a moving average).
- **Stop-loss:** Just below the 61.8% level (or the swing low) — beyond it the trend is in doubt.
- **Target / when to sell:** The prior high (0% level); then Fibonacci EXTENSIONS at 1.272× and 1.618× for runners.

## Pivot points




**What it is:** Support (S1, S2) and resistance (R1, R2) levels auto-calculated from the prior day's high, low and close, around a central pivot.

**Why it works:** They give objective intraday reference levels derived from where the market just agreed value was.

**The psychology:** Because day-traders and algos all compute the same pivots, orders cluster at them — they become self-fulfilling intraday magnets and barriers. Price above the central pivot = bullish bias for the session.

**How to use / trade it:** Use them for intraday targets and stops on indices and F&O; trade bounces/rejections at S/R levels and breakouts through them.

**Every part, explained simply:** Pivot points are the day's support/resistance levels, computed from YESTERDAY's high, low and close. The central **Pivot (P)** is the day's 'fair value'; **R1, R2** are resistances above; **S1, S2** are supports below.

**How it's calculated (formula):** `P = (High + Low + Close) ÷ 3`; `R1 = 2P − Low`; `S1 = 2P − High`; `R2 = P + (High − Low)`; `S2 = P − (High − Low)`.

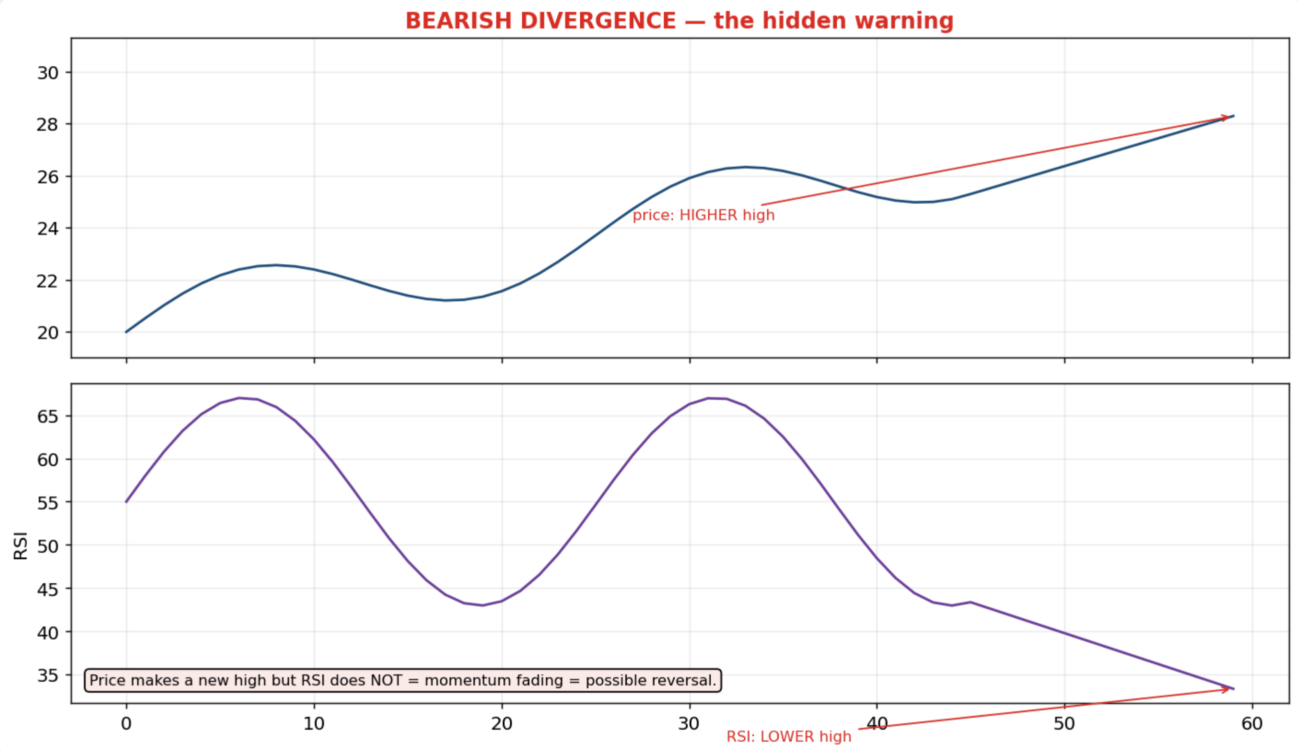
**Worked example:** Yesterday H = 110, L = 100, C = 105 → P = 105, R1 = 2×105−100 = **110**, S1 = 2×105−110 = **100**, R2 = 115, S2 = 95.

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** Intraday with an up-bias: buy near the central Pivot or S1 when it holds.
- **Stop-loss:** Below the next pivot level down.
- **Target / when to sell:** R1 first, then R2; sell / scale out at the resistance pivots.

## PART 4 — Theories

### Divergence — the hidden warning




**What it is:** Price and a momentum indicator (RSI/MACD) disagree — e.g., price makes a higher high but the indicator makes a lower high.

**Why it works:** Price can keep rising on diminishing participation; the indicator exposes that the engine is weakening before the car stops.

**The psychology:** A new price high on weaker momentum means the rally is being carried by fewer, later, and smaller buyers — often smart money is quietly distributing to an euphoric crowd. The crowd sees a new high and feels safe; the divergence reveals the opposite.

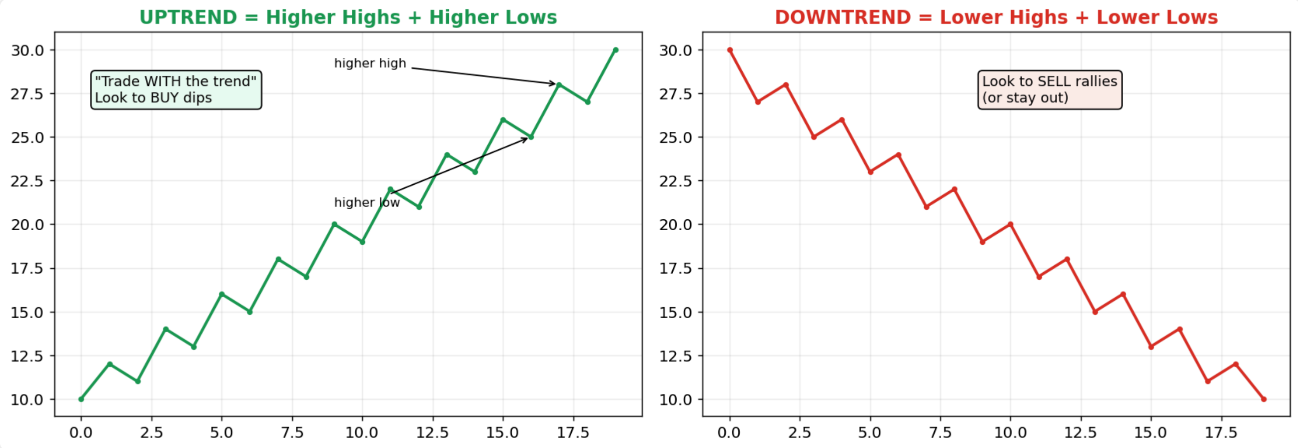
**How to use / trade it:** Treat divergence as an early-warning to tighten stops or look for a reversal trigger — not a stand-alone sell; wait for price confirmation (a break of support).

**Interview depth:** Distinguish regular divergence (warns of reversal) from hidden divergence (signals trend continuation) — mentioning both shows real depth.

> **Every part, explained simply:** Same idea as RSI divergence but it applies to any momentum indicator (RSI or MACD). Compare the indicator's peaks/troughs to price's: if price makes a new extreme that the indicator does NOT match, the move is losing power. **Regular** divergence warns of a reversal; **hidden** divergence (price higher low, indicator lower low in an uptrend) signals the trend will continue.

> **Trade plan — where to buy, stop and sell:**
> - **Buy / entry:** Don't buy on divergence alone — wait for price to BREAK its short-term trendline / structure, then enter.
> - **Stop-loss:** Beyond the divergence extreme (the price high or low that created it).
> - **Target / when to sell:** The prior support/resistance; divergence reversals can run far, so trail the stop.

## Trend structure




**What it is:** Uptrend = higher highs and higher lows; downtrend = lower highs and lower lows; sideways = a range.

**Why it works:** Trends persist because of feedback loops — rising prices attract more buyers and validate holders, which pushes prices higher still.

**The psychology:** An uptrend is a visible story of growing optimism: each dip is bought earlier (higher lows = impatient demand) and each push reaches further (higher highs). The trend ends when that optimism breaks — the first lower high/low signals the psychology has shifted.

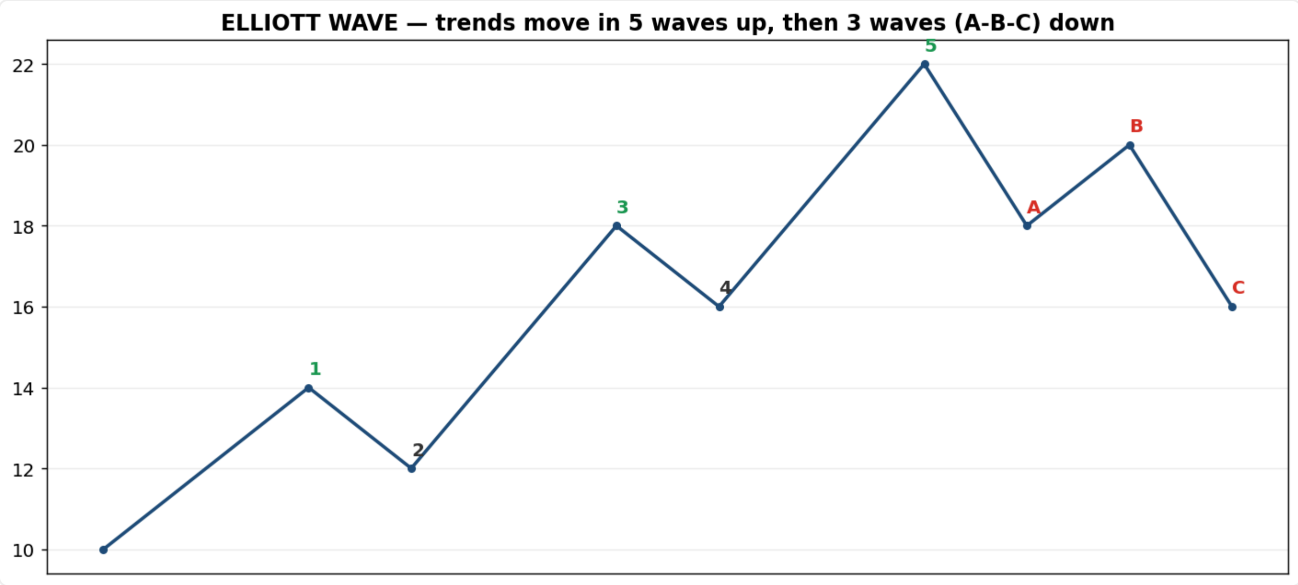
**How to use / trade it:** Always define the trend first and trade with it: buy dips in uptrends, sell rallies in downtrends, stand aside in choppy ranges.

**Every part, explained simply:** Ignore the small wiggles and look at the staircase. **Uptrend** = each high is higher than the last (higher highs) and each dip stops higher than the last (higher lows). **Downtrend** = lower highs and lower lows. **Sideways** = a flat range. The trend 'breaks' the first time the staircase fails — e.g., in an uptrend, the first LOWER low.

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** Buy dips toward the higher lows in an uptrend; sell rallies into lower highs in a downtrend.
- **Stop-loss:** Below the most recent higher low (in an uptrend).
- **Target / when to sell:** The next higher high; trail to ride the trend. EXIT when structure breaks (first lower low).

## Elliott Wave




**What it is:** Trends unfold in 5 waves in the trend direction (1-3-5 up, 2-4 pullbacks), then correct in 3 waves (A-B-C).
**Why it works:** It formalises the natural swing of crowd emotion between optimism and pessimism, and is fractal (the same shape on every timeframe).
**The psychology:** Wave 1 = a few early believers. Wave 2 = doubt (pullback). Wave 3 = the crowd recognises the trend — the strongest, longest wave (peak optimism building). Wave 4 = profit-taking. Wave 5 = euphoria/exhaustion, often on divergence. Then A-B-C unwinds the excess as reality sets in.
**How to use / trade it:** Use it to gauge where a move is in its life-cycle — e.g., be cautious chasing a 'wave 5' that shows momentum divergence. Acknowledge it's subjective and best combined with other tools.

**Every part, explained simply:** Developed by Ralph Nelson Elliott (1930s): prices move in repeating **8-wave cycles (a 5-3 structure)** driven by crowd psychology.
**The 5-wave Motive (Impulse) phase** — moves WITH the main trend:

- **Wave 1** — first move up, a few early buyers.
- **Wave 2** — pullback as early buyers take profit, but it does NOT fall below the start of Wave 1.
- **Wave 3** — usually the longest, strongest wave; good news pulls the public in and price surges.
- **Wave 4** — a mild profit-taking dip that does NOT enter Wave 1's price territory.
- **Wave 5** — the final push; optimism is extreme and the asset is often overvalued (watch for divergence here).

**The 3-wave Corrective phase** — moves AGAINST the trend (a breather), labelled A-B-C:

- **Wave A** — first move against the trend; many mistake it for a normal dip.
- **Wave B** — a bounce that fools people into thinking the old trend is back.
- **Wave C** — a sharp drop that confirms the correction.

**Core ideas:** waves are **fractal** (zoom in and a small 8-wave cycle sits inside a bigger wave, like nested Russian dolls); practitioners use **Fibonacci ratios** (38.2%, 50%, 61.8%) to estimate how deep corrections go and how far impulse waves run.
**Three rules that MUST hold for a valid count:** (1) Wave 2 never retraces more than 100% of Wave 1; (2) Wave 3 is never the shortest of waves 1, 3 and 5; (3) Wave 4 never overlaps Wave 1's price territory.
Because counting waves is subjective, always confirm with other tools (moving averages, momentum oscillators).

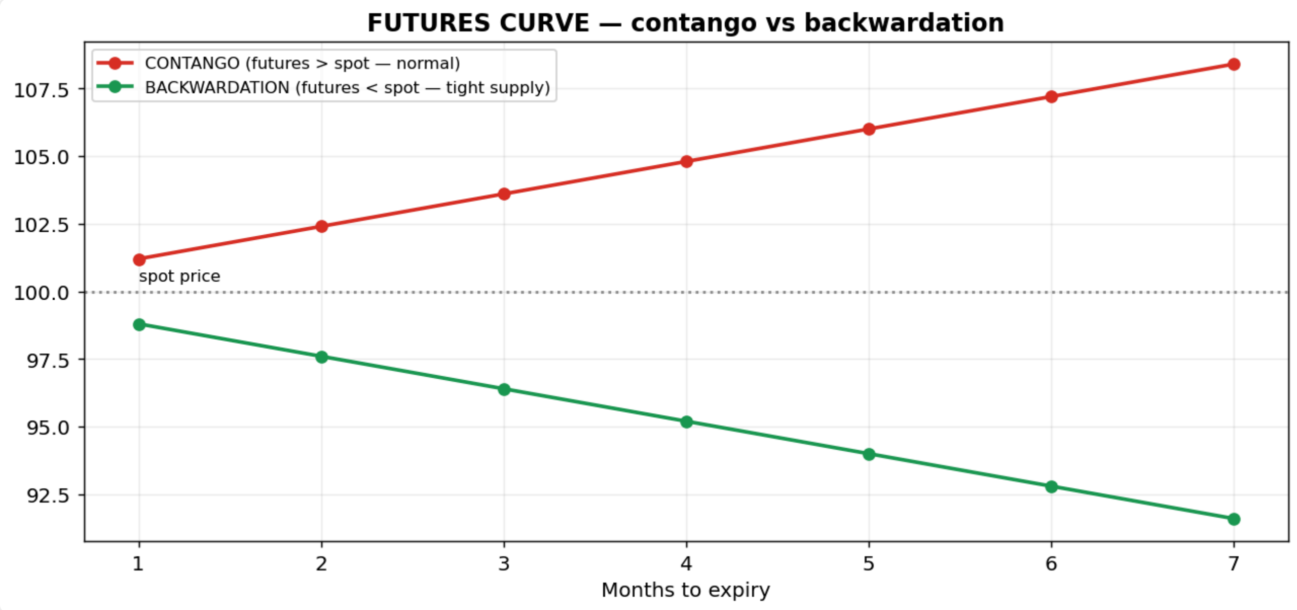
**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** The best entry is the start of wave 3 (right after the wave-2 pullback) — the longest, strongest wave.

- **Stop-loss:** Below the wave-2 low.
- **Target / when to sell:** Wave 3 often extends to 1.618× wave 1; take profit / tighten in wave 5 (exhaustion).

# PART 5 — Futures & Options (F&O)

## Futures: contango vs backwardation




**What it is:** The shape of futures prices across expiries — contango (futures above spot) or backwardation (futures below spot).

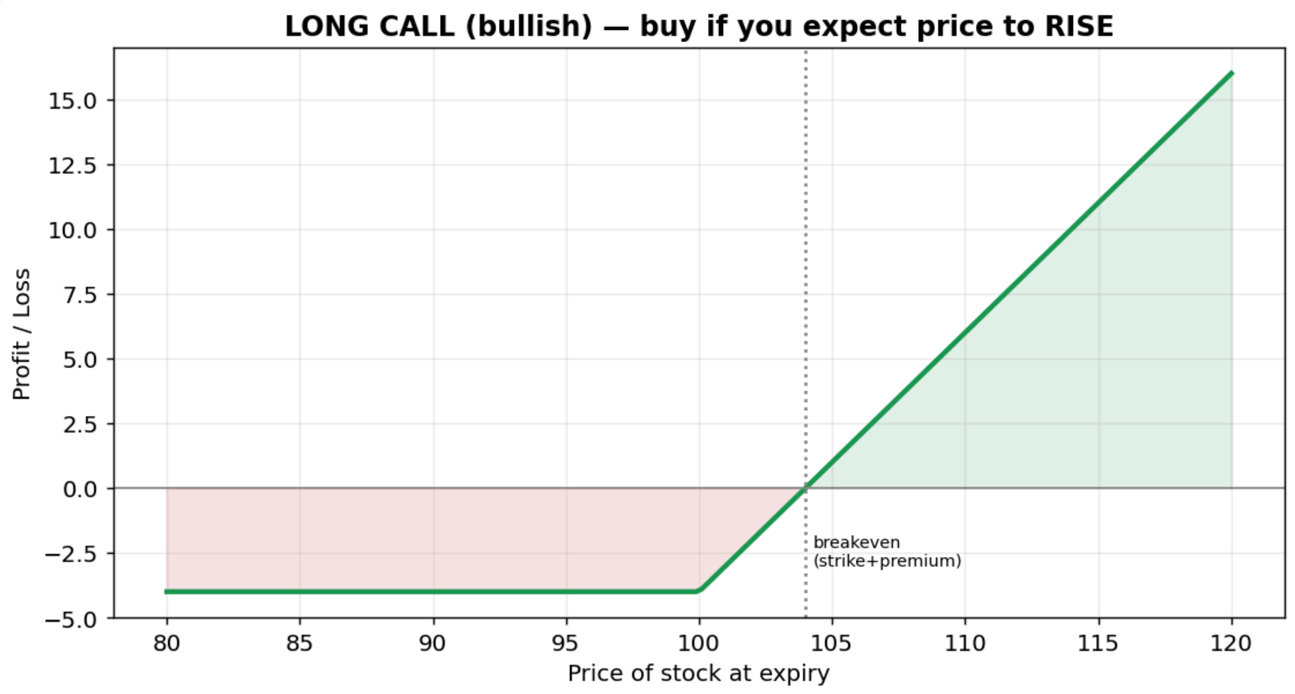
**Why it works:** The curve reflects the cost of carrying the asset versus how urgently the market wants it now.

**The psychology:** Contango is 'normal' — futures cost more because of interest/storage (cost of carry). Backwardation signals near-term scarcity or strong immediate demand — buyers will pay a premium for the asset NOW rather than later (common in tight crude/commodity markets), a sign of bullish urgency.

**How to use / trade it:** Watch the curve and rollovers: heavy long rollover to the next expiry signals trend conviction; backwardation often accompanies strong spot demand.

> **Every part, explained simply: Spot price** = the price if you buy TODAY (immediate delivery). **Futures price** = a price you fix today to buy/sell on a future date (a futures contract).
> **Contango (the normal market):** the futures price is HIGHER than spot. Why? Holding an asset until later has costs — interest on money, storage, insurance — together called the **cost of carry**. The futures curve slopes UP. Memory: **Future > Spot = Contango**.
> **Backwardation (unusual):** the futures price is LOWER than spot. This happens with strong IMMEDIATE demand or a shortage — everyone wants the asset NOW, so today's price is bid up above the future. The curve slopes DOWN. Memory: **Spot > Future = Backwardation** (think of one bottle of water in a desert: huge demand today, but rain may come in 3 months).
> **Where you see backwardation:** mostly commodities — crude oil, natural gas, wheat, sometimes gold — because real shortages happen.
> **Trading meaning:** contango = normal, nothing special; backwardation = strong current demand / tight supply, often a bullish or shortage signal.
> **30-second interview answer:** 'Contango is when the futures price is above spot, usually due to the cost of carry (interest, storage, insurance) — the normal state. Backwardation is the opposite, futures below spot, typically from strong immediate demand or a shortage. The relationship between spot and futures prices across expiries is the futures curve.'
> **How it's calculated (formula):** `Futures Price = Spot Price + Cost of Carry` (cost of carry = interest + storage + insurance − any income such as dividends).
> **Worked example:** Spot ₹100, interest ₹2, storage ₹1, insurance ₹1 → Future = 100+2+1+1 = **₹104** (contango). But if crude turns suddenly scarce and everyone needs it now: Spot ₹100, 3-month Future ₹97 → Future < Spot = **backwardation**.

## Long Call — bet that price will RISE




**What it is:** A call gives you the right (not obligation) to BUY at a fixed strike price. You pay a premium; that premium is your maximum loss.

**Why it works:** It gives leveraged upside with limited, known risk — a small premium controls a large position.

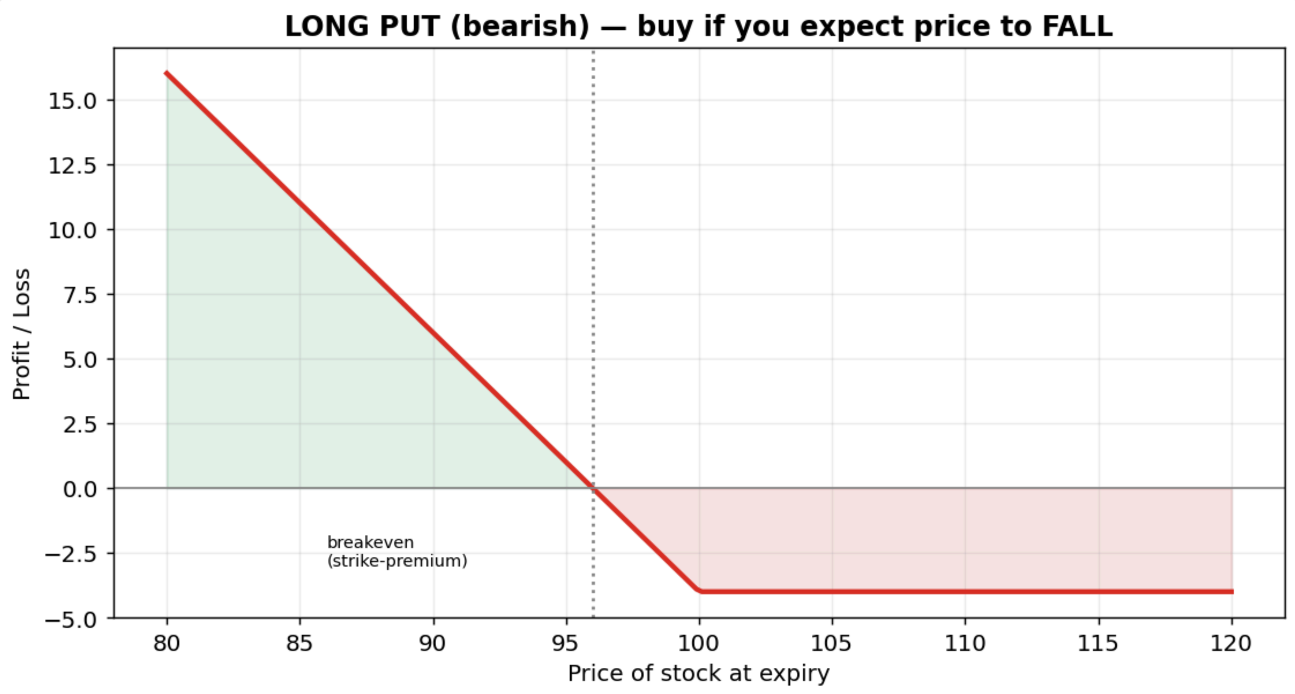
**The psychology:** The buyer pays a little for a big potential payoff (asymmetric) — attractive when you're confident price will rise. The seller collects the premium and bets it won't.

**How to use / trade it:** BUY a call when bullish and your setup triggers; exit at your target or cut if the underlying breaks your stop. Time decay (theta) eats value daily, so don't overstay.

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** Buy a call when your bullish setup triggers.
- **Stop-loss:** Exit if the underlying breaks your stop, or cut at ~30-50% premium loss.
- **Target / when to sell:** Book profit at the underlying's target; don't ride it into expiry (theta decay).

## Long Put — bet that price will FALL




**What it is:** A put gives you the right to SELL at a fixed strike price. You pay a premium; that premium is your maximum loss.

**Why it works:** It profits as price falls, with limited risk — a way to go bearish or to hedge a holding.

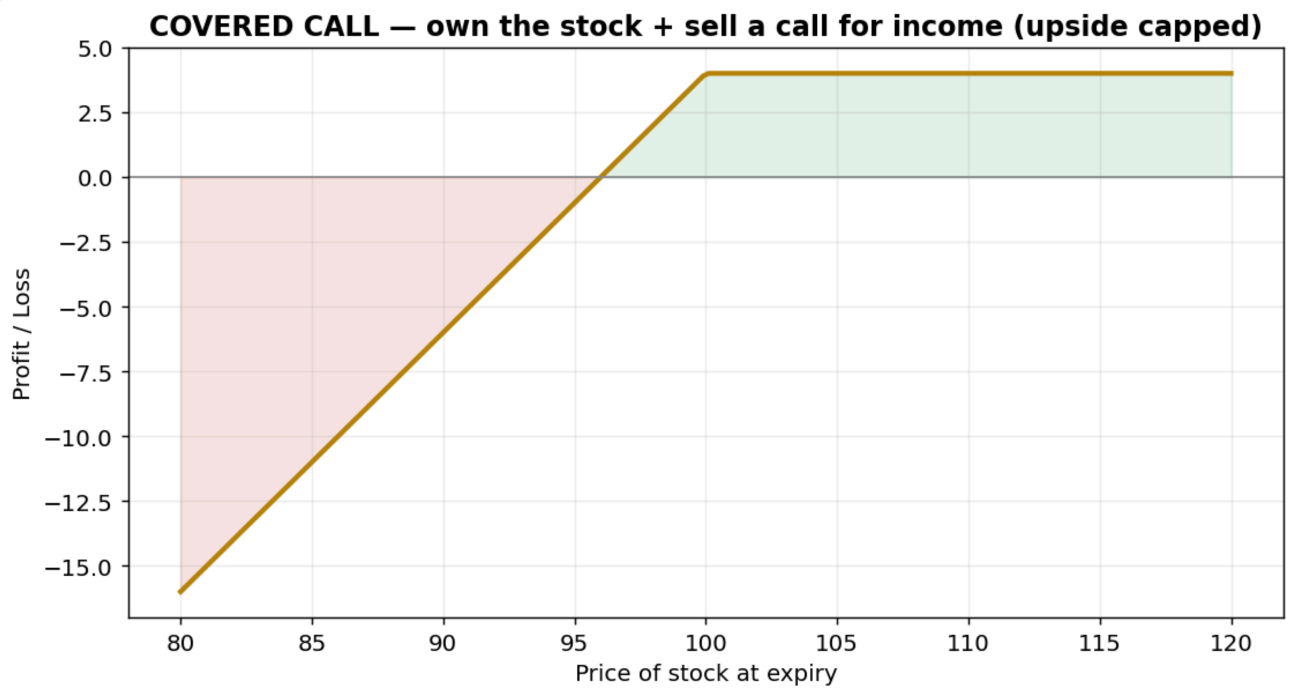
**The psychology:** The buyer pays a small premium for large downside profit; it's also 'insurance' for someone who owns the stock and fears a drop.

**How to use / trade it:** BUY a put when bearish, or to protect a holding; exit at target or if the underlying reclaims your level. Theta works against you each day.

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** Buy a put when bearish, or to hedge a stock you own.
- **Stop-loss:** Exit if the underlying reclaims your level.
- **Target / when to sell:** Book profit at the target; mind daily theta decay.

## Covered Call — earn income on a holding

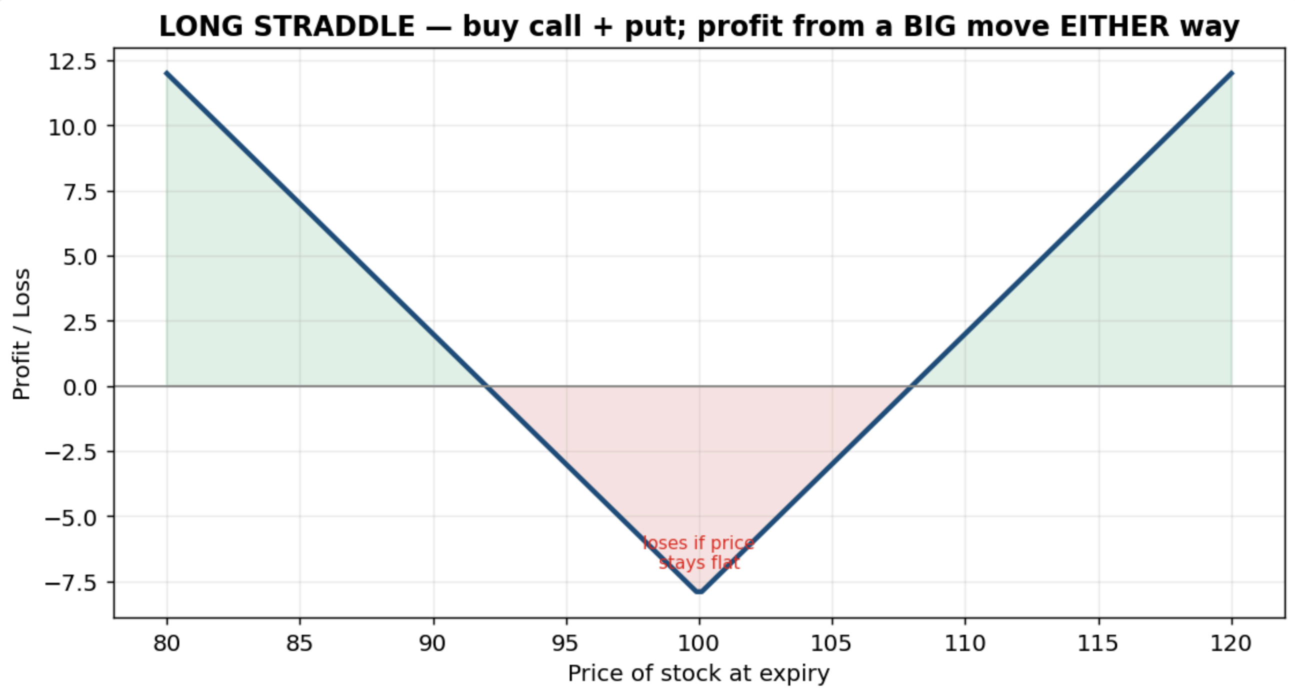



**What it is:** You OWN the stock and SELL a call against it, collecting the premium as income.
**Why it works:** It turns a flat/mildly-bullish holding into income, in exchange for capping your upside.
**The psychology:** A mildly bullish, income-seeking mindset: you're happy to give up big upside for a steady premium while you hold the stock.
**How to use / trade it:** Use it on stocks you already own and expect to move sideways; you keep the premium if price stays below the strike, but your gains are capped above it.

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** Sell a call against stock you already own, in a sideways/mildly-bullish view.
- **Stop-loss:** The real risk is the stock falling (premium gives a small cushion).
- **Target / when to sell:** Keep the premium if price stays below the strike; gains are capped above it.

## Long Straddle — bet on a BIG move either way




**What it is:** Buy a call AND a put at the SAME strike. You profit if price moves sharply in EITHER direction.

**Why it works:** It's a pure volatility bet — direction doesn't matter, only the size of the move.

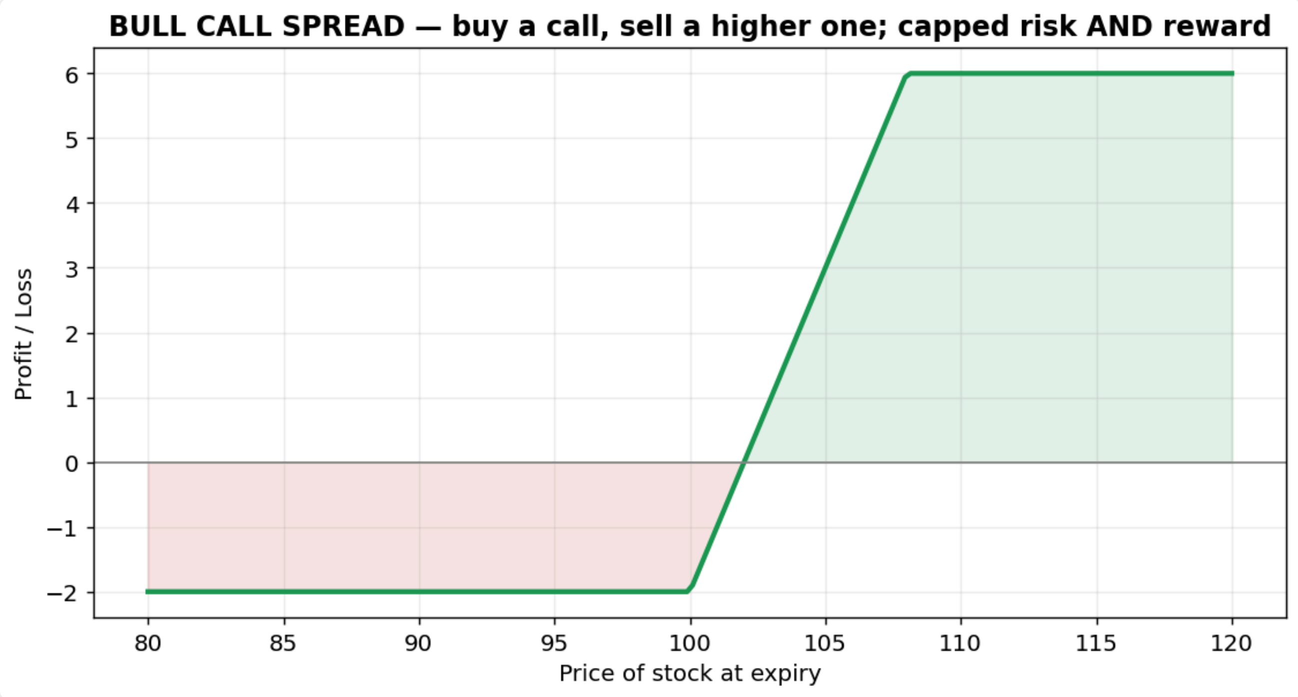
**The psychology:** The buyer doesn't care up or down, only that something big happens — typically bought before events (results, budget) when a large move is expected but the direction is unknown.

**How to use / trade it:** BUY before a known event; you need a move bigger than the combined premium to profit. Exit fast after the event — volatility (IV) collapses and theta accelerates.

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** Buy before a known event (results/budget) when a big move is expected.
- **Stop-loss:** Time-based: exit if the move doesn't arrive, before the post-event IV crush.
- **Target / when to sell:** Take profit once the move exceeds the combined premium; exit quickly after the event.

## Bull Call Spread — cheaper bullish bet




**What it is:** Buy a call AND sell a higher-strike call. This caps both your cost and your reward.

**Why it works:** It expresses a moderate bullish view for less money than a naked call, with a known maximum loss.

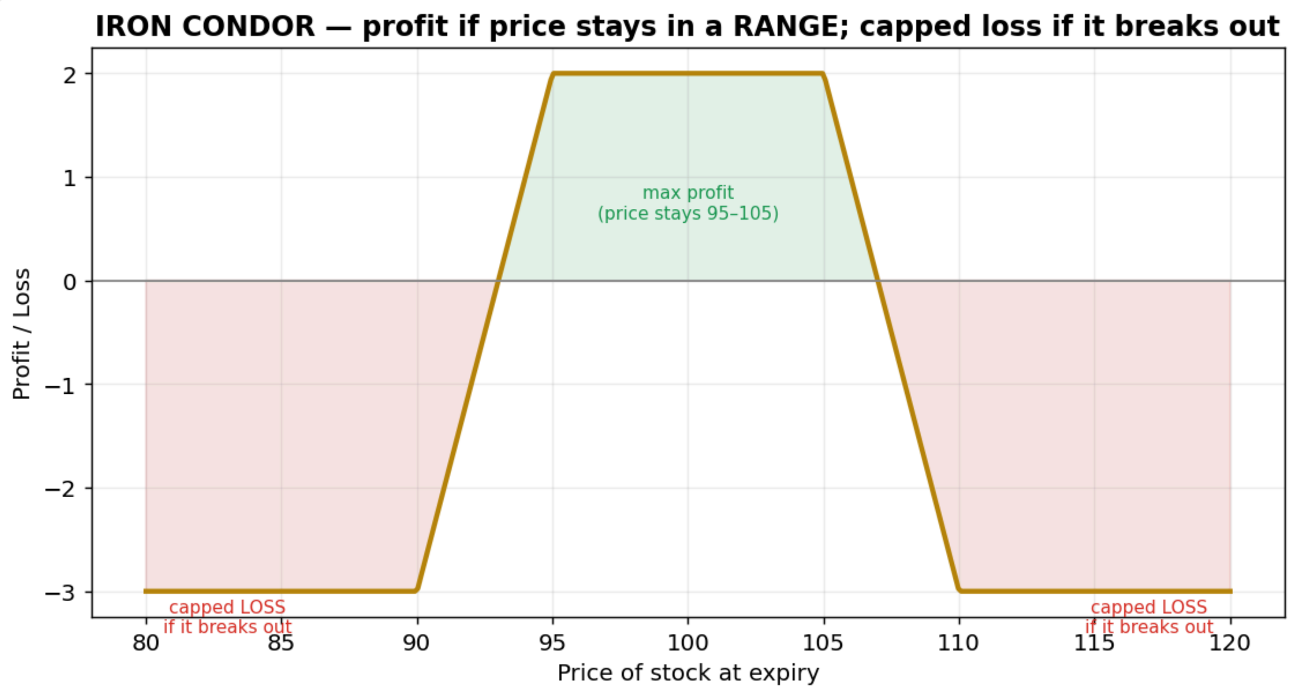
**The psychology:** A disciplined, probability-minded approach: you accept a capped reward in exchange for lower cost and defined risk.

**How to use / trade it:** Use when you expect a moderate rise (not a moonshot); max profit near the higher strike, max loss = the net premium paid.

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** Enter for a moderate bullish view (cheaper than a naked call).
- **Stop-loss:** Max loss is fixed = the net premium paid.
- **Target / when to sell:** Max profit near the higher (sold) strike.

## Iron Condor — profit when price goes NOWHERE




**What it is:** Sell an out-of-the-money call spread AND put spread. You profit if price stays in a range.

**Why it works:** It monetises a calm, range-bound market by collecting premium that decays with time.

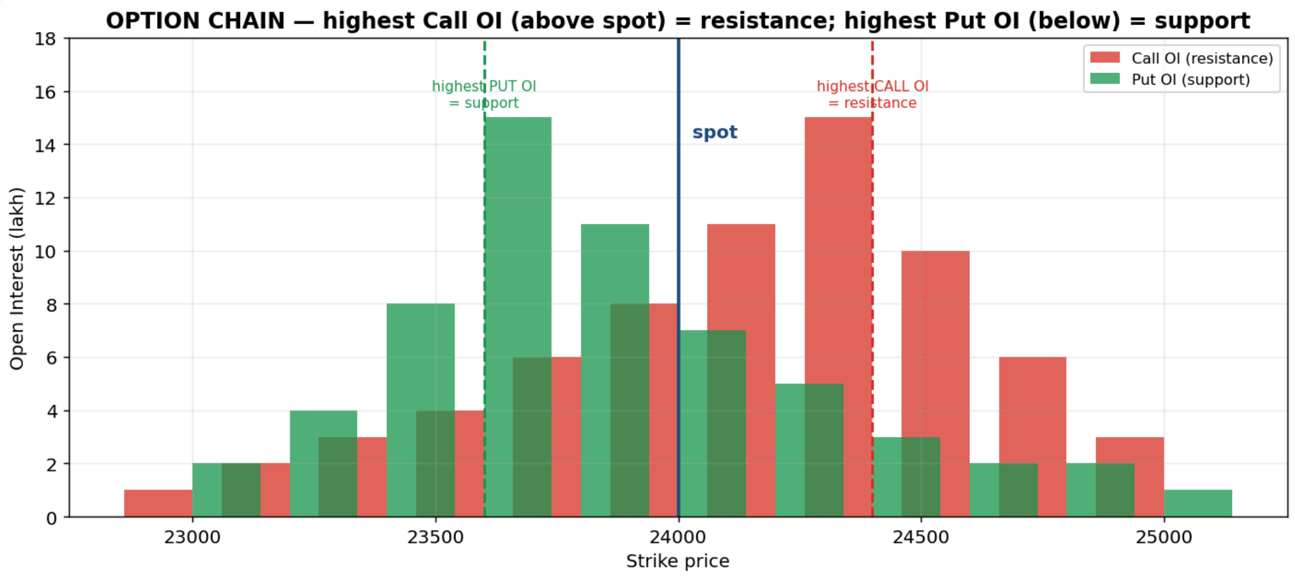
**The psychology:** The seller is betting on quiet markets and harvests time decay as long as price stays inside the range — effectively selling 'insurance' to nervous traders.

**How to use / trade it:** Use when you expect low volatility; close at ~50% of max profit rather than holding to expiry. The danger is a sudden breakout beyond your short strikes.

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** Enter when you expect a quiet, range-bound market.
- **Stop-loss:** Act if price approaches one of your short strikes.
- **Target / when to sell:** Close at about 50% of max profit rather than holding to expiry.

## Open Interest & the option chain




**What it is:** Open Interest = number of outstanding contracts at each strike, displayed across the option chain.

**Why it works:** It shows where large positions (often institutional writers) are concentrated, which creates magnetic support/resistance.

**The psychology:** The strike with the highest CALL open interest tends to act as resistance — call writers (who collected premium) defend it and want price to stay below. Highest PUT open interest acts as support for the same reason. **Max pain** is the strike where the most options expire worthless, and price often drifts there near expiry because large writers are incentivised to push it.

**How to use / trade it:** Read OI to map likely intraday/expiry support and resistance; rising OI + rising price = fresh longs (strong), rising OI + falling price = fresh shorts.

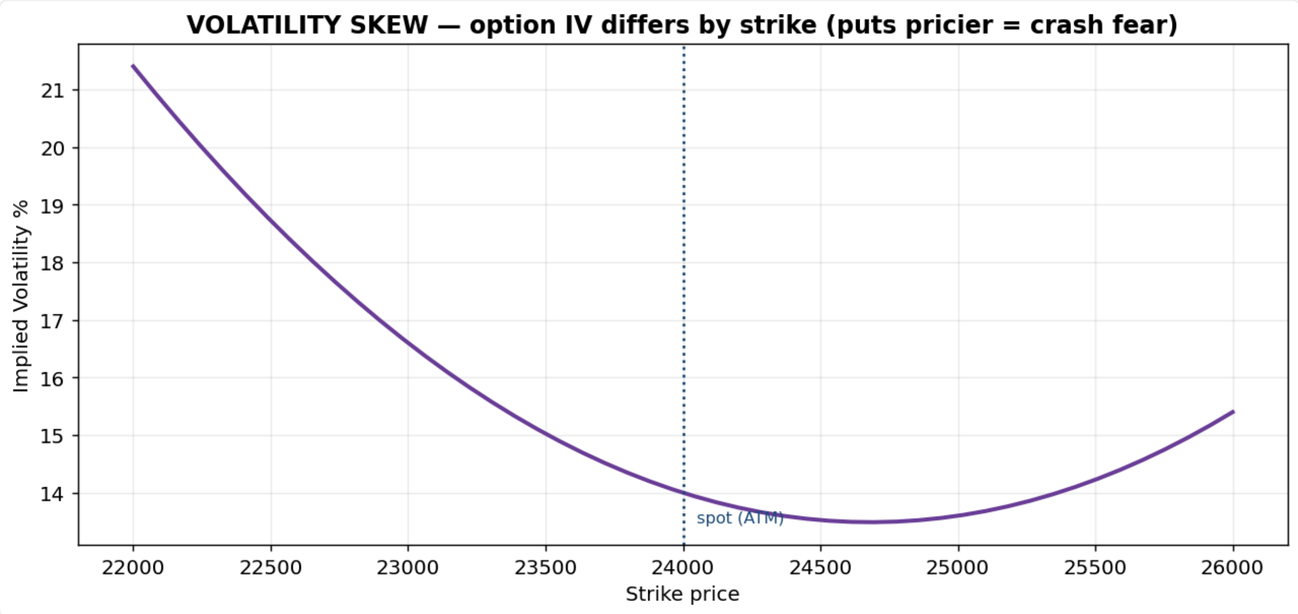
**Interview depth:** Mention PCR (put-call ratio) as a sentiment gauge — a very high PCR can be contrarian bullish (excessive fear).

> **Every part, explained simply: Open Interest (OI)** = the number of option contracts still open at each strike. Big OI = a lot of money parked there. The strike with the highest **call** OI acts as resistance (call sellers defend it); the highest **put** OI acts as support. **PCR** (put-call ratio) = total put OI ÷ call OI — a sentiment gauge; very high PCR can be contrarian bullish.

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** Treat the highest-PUT-OI strike as support (buy bounces near it) and the highest-CALL-OI strike as resistance (sell/short into it).
- **Stop-loss:** Just beyond the OI 'wall' you are trading against.
- **Target / when to sell:** The opposite OI wall; near expiry price often gravitates toward the max-pain strike.

## Implied Volatility & skew



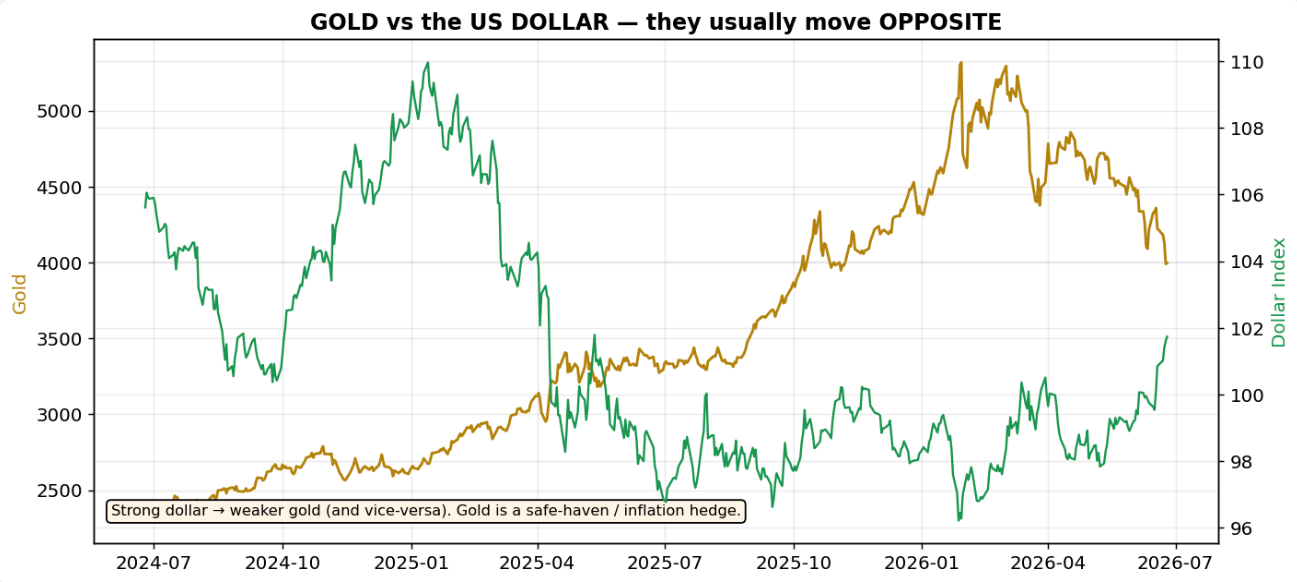

**What it is:** IV is the market's expected future volatility implied by option prices; skew = IV differing across strikes.
**Why it works:** Option prices reveal what the crowd expects volatility to be — fear and demand for protection get priced in.
**The psychology:** Downside puts usually carry higher IV (the 'skew') because investors fear crashes more than melt-ups and pay up for protection — a measurable fingerprint of fear. High overall IV means options are expensive (lots of expected movement / anxiety); low IV means complacency.
**How to use / trade it:** Buy options when IV is low and sell premium when IV is high; reading the IV surface (across strikes and expiries) is exactly the volatility-analysis work you did on the desk.

> **Every part, explained simply: Implied Volatility (IV)** = the market's expectation of future movement, baked into option prices — high IV = expensive options (big moves expected), low IV = cheap (calm expected). The **skew** means IV differs by strike: downside puts usually cost more because investors fear crashes and pay up for protection. There's no neat formula — IV is solved by reversing the Black-Scholes pricing model.

> **Trade plan — where to buy, stop and sell:**
> - **Buy / entry:** BUY options (calls/puts) when IV is LOW (they're cheap); SELL premium (spreads/condors) when IV is HIGH.
> - **Stop-loss:** Manage by the underlying's price level, as in any trade.
> - **Target / when to sell:** For long options, exit BEFORE an expected IV crush (e.g. right after a results announcement).

## PART 6 & 7 — Commodities, Indices & Sentiment

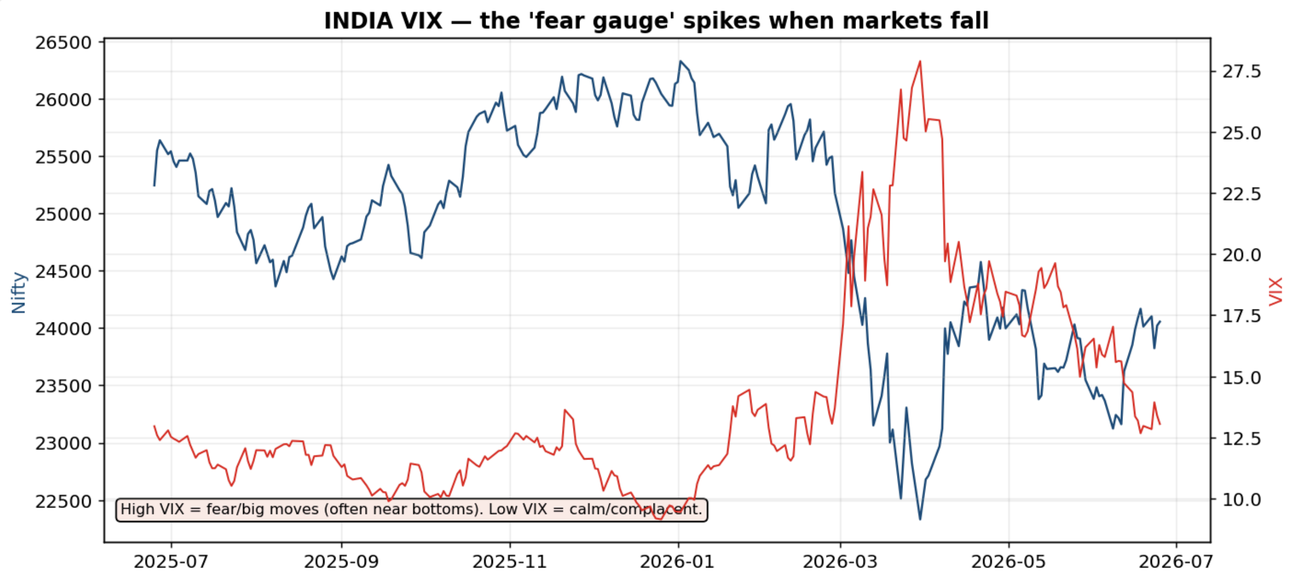
### Gold vs the US Dollar




**What it is:** Gold (priced in USD) and the US Dollar Index usually move in opposite directions.
**Why it works:** Because gold is priced in dollars and competes with the dollar as a store of value, a stronger dollar mechanically and psychologically weakens gold.
**The psychology:** Gold pays no interest, so when real yields/the dollar rise, money prefers cash/bonds and gold falls. In fear or high inflation, investors flee to gold as a safe haven and inflation hedge — gold rises on anxiety. It's a barometer of distrust in paper money.
**How to use / trade it:** Track the dollar, real yields and risk sentiment when forming a gold view; crude instead keys off OPEC supply, global demand and geopolitics.

**Trade plan — where to buy, stop and sell:**
- **Buy / entry:** Lean LONG gold when the dollar / real yields are falling or fear is rising; lean SHORT when the dollar strengthens.
- **Stop-loss:** Below gold's nearest support level.
- **Target / when to sell:** The next resistance — confirm with the usual technicals (trend, RSI, levels).

## India VIX — the fear gauge




**What it is:** VIX measures the market's expected 30-day volatility, derived from index option prices.

**Why it works:** When traders expect big moves they pay up for options, which pushes VIX up — so it reads collective anxiety.

**The psychology:** VIX spikes during selloffs because fear drives a rush to buy protective puts. Extreme VIX often coincides with capitulation bottoms (peak fear = sellers exhausted); very low VIX signals complacency and can precede sharp shocks.

**How to use / trade it:** Use VIX as a contrarian sentiment tool — spikes near support can flag bounce opportunities; persistently low VIX warns against complacency.

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** A VIX spike to an extreme near a major support level = look to BUY the index (fear is peaking).
- **Stop-loss:** Below that support level.
- **Target / when to sell:** A mean-reversion bounce; trim the position as VIX falls back toward normal.

## PART 8 — Risk Management (what makes a pro)

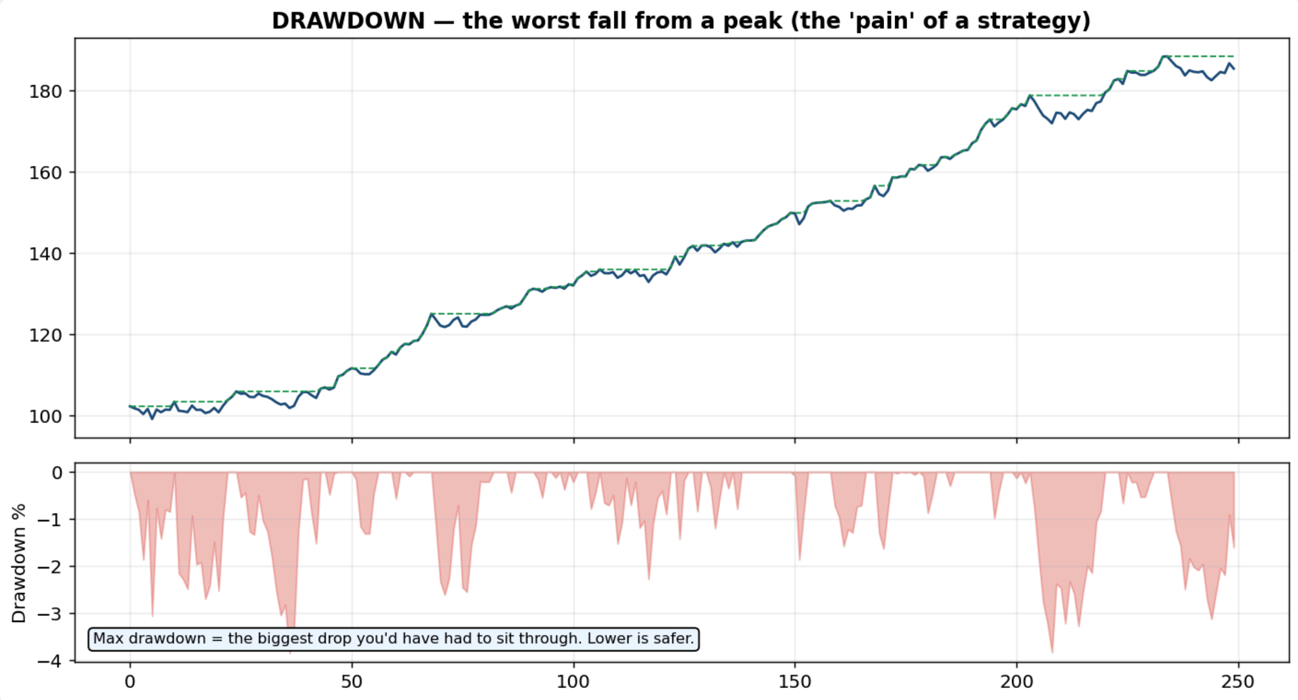
### Risk : Reward and stops


**What it is:** Every trade defined by an entry, a stop-loss (exit if wrong) and a target, with reward sized larger than risk.
**Why it works:** Positive expectancy doesn't require being right often — it requires winners bigger than losers.
**The psychology:** The stop-loss exists to defeat loss aversion — the human urge to hold losers hoping they recover, which turns small losses into account-killers. Pre-deciding the exit removes emotion in the heat of the moment. A 1:2+ reward:risk means you can be wrong more than half the time and still grow capital.
**How to use / trade it:** Demand at least 2:1 reward:risk, set the stop at a level that invalidates your idea (not an arbitrary %), and never move a stop further away to avoid being wrong.
**Interview depth:** Frame it as 'survive first': controlling the downside is what keeps you in the game long enough for your edge to play out.

**Every part, explained simply:** Three prices define every trade: **Entry** (where you buy), **Stop-loss** (exit if wrong), **Target** (take profit). Reward = entry-to-target distance; Risk = entry-to-stop distance.
**How it's calculated (formula):** `Reward : Risk = (Target − Entry) ÷ (Entry − Stop)`.
**Worked example:** Buy at 100, target 112, stop 96 → reward = 12, risk = 4 → ratio = **3 : 1**. At 3:1 you can be wrong on 2 of every 3 trades and still break even.

**Trade plan — where to buy, stop and sell:**
- **Buy / entry:** Only take the trade if, from your entry, the distance to a sensible target is at least 2× the distance to your stop.
- **Stop-loss:** At the price that proves your idea WRONG (below support/structure) — never an arbitrary percentage.
- **Target / when to sell:** At least 2× your risk; scale out part at 1× and trail the rest.

## Drawdown — the pain measure




**What it is:** Drawdown is the fall from an equity peak; max drawdown is the worst such fall a strategy/account suffered.

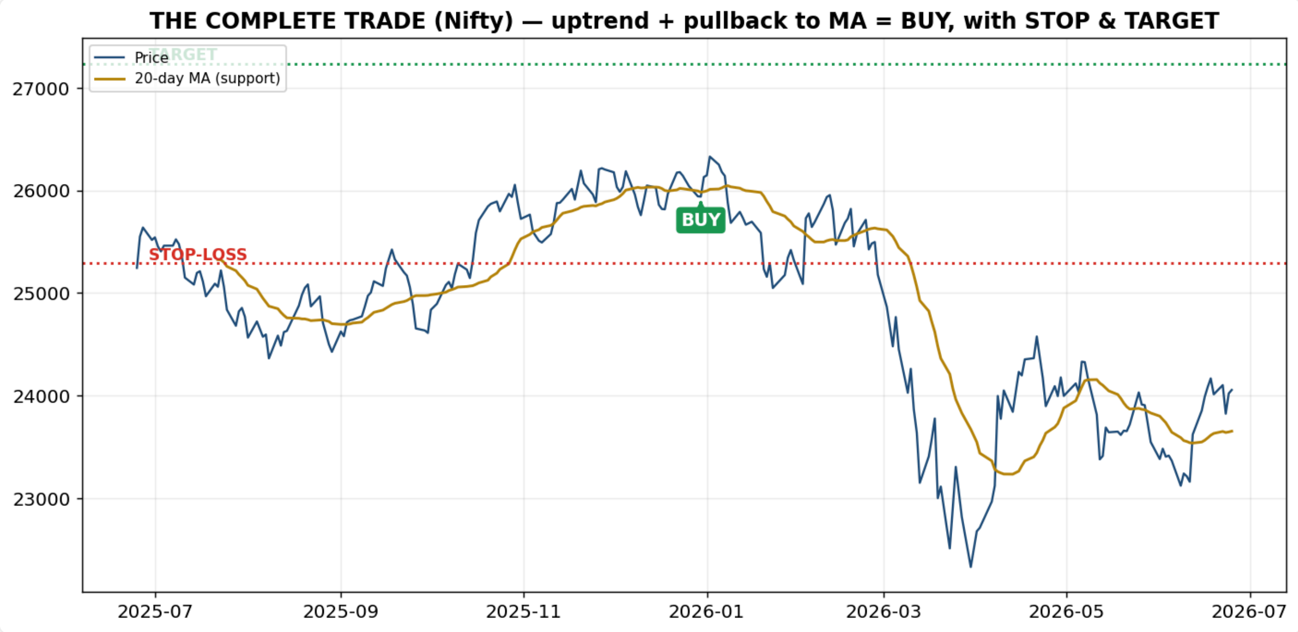
**Why it works:** Returns alone are meaningless without knowing the pain endured to get them — drawdown measures risk of ruin and of quitting.

**The psychology:** Big drawdowns break traders psychologically — most abandon a strategy at its worst point, locking in the loss. A strategy with lower drawdown is one you can actually stick with, which is why risk-adjusted measures (Sharpe, Sortino) matter more than raw return.

**How to use / trade it:** Judge any strategy by drawdown and Sharpe/Sortino, not just total return; size positions so a normal drawdown is survivable both financially and emotionally.

> **Every part, explained simply:** Drawdown is how far your account has fallen from its highest point (peak). Max drawdown = the worst such fall — the most pain you'd have had to sit through.
> **How it's calculated (formula):** `Drawdown = (Peak − Trough) ÷ Peak`.
> **Worked example:** If equity went 100 → 150 → 90, the peak was 150 and trough 90 → drawdown = (150−90)/150 = **40%**.

## Putting it ALL together — a real trade




**What it is:** A complete setup: confirm the uptrend, wait for a pullback to support/MA, then buy with a stop just below and a target above.

**Why it works:** It stacks independent edges — trend, level, momentum, and favourable risk:reward — so the probabilities line up in your favour.

**The psychology:** You're buying where the crowd that missed the move re-enters (support), in the direction the dominant players favour (trend), with a pre-planned exit that keeps emotion out. Each confluence factor is another group of traders acting in your favour.

**How to use / trade it:** Checklist every trade: trend ✓ → level ✓ → trigger (candle/indicator) ✓ → entry, stop, target with reward > risk. If any piece is missing, pass.

**Trade plan — where to buy, stop and sell:**

- **Buy / entry:** Uptrend confirmed + pullback to support/MA + a bullish trigger candle → buy.
- **Stop-loss:** Just below that support / moving average.
- **Target / when to sell:** The next resistance or a measured move, sized for 2:1+; trail the stop to ride the trend.

# PART 9 — More Market Theory (Wyckoff, Sector Rotation, Breadth)

## Wyckoff Method (in depth)

Richard Wyckoff (early 1900s) studied how **large operators ("smart money") accumulate and distribute** without moving price against themselves. He imagined all market activity as one big operator — the **"Composite Man"** — who buys low from the fearful public, marks price up, sells high to the greedy public, then marks it down. If you learn to read his footprints, you can position alongside him.

**The three Wyckoff laws (say these in an interview):** 1. **Supply & Demand** — price rises when demand beats supply, falls when supply beats demand. Everything reduces to this. 2. **Cause & Effect** — the size of a sideways accumulation/distribution range (the "cause") sets the size of the move that follows (the "effect"). A long base → a big move. 3. **Effort vs Result** — compare volume (effort) to price progress (result). Heavy volume but little price movement = **absorption** (smart money quietly soaking up shares) and warns of a turn.

**The full market cycle: Accumulation → Mark-up → Distribution → Mark-down.**

**Accumulation range — the key events (bottoming):** - **PS** (Preliminary Support) and **SC** (Selling Climax) — panic selling on huge volume marks the low. - **AR** (Automatic Rally) — a sharp bounce as selling dries up; it sets the top of the range. - **ST** (Secondary Test) — price retests the low on lighter volume (sellers exhausted). - **Spring** — a brief false breakdown BELOW support that traps the last sellers, then snaps back up. *The classic Wyckoff buy signal.* - **SOS** (Sign of Strength) — a strong rally with wide spread and volume. - **LPS** (Last Point of Support) — a higher low before the markup begins → entry point.

**Distribution range — the mirror events (topping):** - **PSY** (Preliminary Supply), **BC** (Buying Climax) — euphoric buying on huge volume marks the high. - **AR**, **ST** — bounce and retest. - **UTAD** (Upthrust After Distribution) — a false breakout ABOVE resistance that traps the last buyers, then fails. *The classic Wyckoff sell signal.* - **SOW** (Sign of Weakness), **LPSY** (Last Point of Supply) — a lower high before the markdown.

**How to use it:** find the trading range, then watch for a **spring** (buy) or **upthrust** (sell) with volume confirmation.

## Market cycles & sector rotation (in depth)

The economy moves in a repeating cycle, and **different sectors lead at each stage** because their businesses thrive at different times:

| Cycle stage | What's happening | Leading sectors |
|---|---|---|
| **Early recovery** | Rates low, recovery begins | Financials/Banks, Auto & Consumer Discretionary, Technology |
| **Mid-cycle (growth)** | Strong expansion | Technology, Industrials, Capital Goods |
| **Late-cycle (peak)** | Inflation rising, economy hot | Energy, Metals & Materials, Commodities |
| **Recession (slowdown)** | Growth falling, fear | Defensives — FMCG (staples), Pharma/Healthcare, Utilities |

- **Cyclicals** (banks, autos, metals, realty) swing hard with the economy; **defensives** (FMCG, pharma, utilities) hold up in downturns.
- **Relative Strength (RS)** is the analyst's key tool: plot a sector index ÷ the broad index (e.g. **Nifty IT ÷ Nifty 50**). A **rising RS line = the sector is outperforming** = leadership; a falling RS line = laggard. Money rotates from tiring sectors into leading ones, and a TRA tracks that flow.
- **In India** you watch the NSE sectoral indices: **Bank Nifty, Nifty IT, Auto, FMCG, Pharma, Metal, Energy, Realty, Fin Nifty.**

## Market breadth (in depth)

Breadth measures **participation** — how many stocks are moving, not just where the index closed. A rally carried by a handful of heavyweights is fragile; broad participation is durable.

- **Advance/Decline (A/D) Line** — a running total of (advancing stocks − declining stocks) each day. Rising A/D **confirms** a healthy rally. If the index makes a new high but the A/D line does NOT (a **breadth divergence**), the rally is narrow and likely to fail.
- **A/D Ratio** — advancers ÷ decliners for the day (a quick strength gauge).
- **% of stocks above their 200-day (or 50-day) MA** — above 50% = broadly healthy market; collapsing = weakening.
- **New Highs vs New Lows** — many fresh 52-week highs = strong breadth; expanding new lows = danger.
- **McClellan Oscillator** — a momentum reading of breadth.
- **TRIN (Arms Index)** — blends advancers/decliners with their volume; above 1 = selling pressure, below 1 = buying.
- **Breadth thrust** — a rare surge where a huge share of stocks rally at once; often marks the start of a new bull leg.

**For a TRA:** before trusting a Nifty move, check breadth — is the whole market participating, or just five large stocks?

---

## PART 10 — Writing Research Reports & Making Calls

*This is your actual daily output as a Technical Research Analyst — master it.*

### The report types (cadence)

- **Daily morning note** (before 9:15 am) — overnight global cues, key levels for Nifty/Bank Nifty/commodities, and 1–3 top trade ideas for the day.
- **Weekly report** — review the week's trend, sector leaders/laggards, and positional calls to hold for days/weeks.
- **Monthly report** — the big-picture outlook, dominant themes, and longer-term support/resistance.
- **Flash / intraday note** — a short alert when something big happens (a breakout, an event, a level breached).

### Anatomy of a daily research report (the sections, in order)

1. **Header** — date, your name, "for educational/clientele use," and a disclaimer.
2. **Market overview** — what happened, and the **global cues**: US close (Dow/S&P/Nasdaq), **GIFT Nifty** (signals our open), **crude, the Dollar Index (DXY), US 10-yr yield**, and **FII/DII flows**.
3. **Index view — Nifty & Bank Nifty** — the trend, the key **support and resistance** levels, and a clear **bias** (bullish / bearish / neutral) with the levels that flip it.
4. **Technical read** — what the indicators say (RSI, MACD, moving averages, Bollinger) and, for F&O, the **option-chain signals** (highest call/put OI, PCR, India VIX).
5. **Trade ideas / stock & sector calls** — the specific recommendations.
6. **Risk note & disclaimer.**

### The trade-call format (every call MUST have all of these)

> **Instrument · View (Buy/Sell/Hold) · Entry · Target(s) · Stop-loss · Time horizon · Rationale**

The **rationale** ties together *trend + level + indicator (+ OI/IV for F&O)*. **A call without a stop-loss is unprofessional** — always give the stop and an honest reward:risk (aim ≥ 1:2).

**Worked example of a clean call:**

> **Bank Nifty — BUY** above 58,400 · **Targets** 59,200 / 60,000 · **Stop-loss** 57,600 (closing basis) · **Horizon** 3–5 sessions · **Rationale:** price reclaimed the 20-DMA and broke the descending trendline; MACD turned bullish and highest put OI sits at 58,000 (support). Reward:risk ≈ 1:2. *Invalidation: a close below 57,600 negates the view.*

### How to write it well (the discipline)

- **Be specific with levels** — "buy above 58,400," not "buy on dips."
- **Talk in probabilities, never guarantees** — "favourable risk-reward to go long above X," not "this will go up."
- **Lead with the call, then the reason** — RMs relay your view to non-technical clients, so the action comes first.
- **Always state the invalidation** — the level/condition that proves the idea wrong.
- **Track and own every call** — log entry, exit, and outcome (target hit / stop hit / open). Your **track record is your credibility**.

### Common mistakes to avoid

No stop-loss · vague levels · fighting the bigger trend · over-issuing calls · promising returns · ignoring global cues and breadth.

## Sample reports (in your projects folder)

Your `projects/2_daily_technical_research_report/` already generates live **daily, weekly and monthly** reports ( `research_report_daily.pdf` , `research_report_weekly.pdf` , `research_report_monthly.pdf` ). There is also a **written-prose sample morning note** ( `SAMPLE_Morning_Research_Note.pdf` ) that shows exactly how a human analyst's note reads — use it as your template.

---

# PART 11 — Regulation, NISM & Ethics

*You're NISM-certified — know these basics cold.*

## NISM Series-XV: Research Analyst

The **SEBI-mandated certification** required to legally publish research or buy/sell recommendations in India. It covers technical analysis, fundamentals, valuation, and — importantly — **regulation and ethics**.

## SEBI (Research Analysts) Regulations — key duties

- **Disclosure of conflicts** — reveal any holding or interest in a security you recommend.
- **No front-running** — never trade ahead of your own published research.
- **Separation** of research from the dealing/trading and investment-banking functions.
- **Rational basis & records** — every recommendation must be justified and documented.
- **Fair dealing** — no miseleading claims, no guaranteed returns, suitability in mind.

## Why it matters in interviews

Showing you understand **disclosures, conflict management, and that research must be compliant** signals maturity beyond just chart-reading.

---

# PART 12 — Behavioural Finance & Market Psychology

Markets are driven by **fear and greed** — technical analysis is, at its heart, the study of crowd psychology.

## The emotional cycle of a market

optimism → excitement → **euphoria (the top)** → anxiety → denial → fear → **panic / capitulation (the bottom)** → hope → optimism. Tops form at maximum greed; bottoms at maximum fear.

## Common biases to know (and name)

- **Herd mentality** — following the crowd instead of the analysis.
- **FOMO** (fear of missing out) — chasing a move that's already run.
- **Loss aversion** — holding losers (hoping they recover) but cutting winners early; the single biggest destroyer of accounts.
- **Confirmation bias** — only seeing evidence that supports your existing view.
- **Anchoring** — fixating on a price ("I'll sell when it gets back to ₹X").
- **Recency bias** — over-weighting what just happened.
- **Overconfidence** — taking oversized risk after a few wins.

## The professional edge

It isn't a secret indicator — it's **discipline**: a written plan, predefined stops, and the emotional control to follow them when the crowd panics. That is what separates a professional from a gambler.

# PART 13 — Tools of the Trade & Common Setups

## Tools

- **Charting:** TradingView (industry standard), broker terminals, MetaTrader, and pro terminals like Bloomberg / Refinitiv (Reuters).
- **Data & screening:** NSE/MCX option chains, screeners (e.g. Chartink), and Excel for tracking and models.
- **Coding (your edge):** Python (pandas, yfinance, matplotlib) for backtesting, screening and automating reports — exactly what your projects show.
- **Daily workflow:** pre-market prep → global cues + key levels → intraday monitoring → end-of-day report → tracking your calls.

## Common trading setups (putting it all together)

- **Trend pullback** — in an uptrend, buy a dip to a moving average or support, with the trend.
- **Breakout** — enter when price clears a key level/range on rising volume; stop just inside the range.
- **Reversal** — trade a confirmed reversal pattern + divergence at a major level (higher risk).
- **Mean-reversion** — in a range, fade the extremes (buy oversold support, sell overbought resistance).
- **Momentum** — buy relative-strength leaders; ride until momentum fades.
- **Event/volatility (F&O)** — straddles/strangles around results or events when a big move is expected.

## The analyst's mental checklist (use this on every idea)

1. What's the **trend** on the higher timeframe?
2. Where are the key **support/resistance** levels?
3. What do **price action / candles** say at those levels?
4. Do **indicators** (one trend + one momentum + volume) confirm? Any **divergence**?
5. For F&O: what do **OI, PCR and IV** say?
6. Define the trade: **entry, target, stop, risk:reward, horizon.**
7. **Size** the position to risk only a small fixed %.
8. Write the call **clearly with a rationale** — and **track** it.

## PART 14 — The Story Behind Every Name (memory hooks)

### Candlestick patterns (these are Japanese, ~300 years old — most names describe what the candle LOOKS like)

| Name | Why it's called that — your memory hook |
|---|---|
| **Doji** | Japanese for *"same time."* The open and close happen at the **same price** → almost no body. Hook: *open = close = doji.* |
| **Hammer** | It **looks like a hammer** — small head (body) on top, long handle (lower wick). It "hammers out the bottom" of a fall. Hook: *a hammer bangs in the floor.* |
| **Hanging Man** | Same shape as a hammer but at a **top** — looks like a man hanging, body = head, long lower wick = dangling legs. Appears after a rise = danger. |
| **Shooting Star** | Small body at the bottom, long upper wick — looks like a **star that shot up and is falling back down.** Appears at tops. Hook: *a falling star = price about to fall.* |
| **Marubozu** | Japanese for *"bald / shaven head"* — a full body with **no wicks (no hair/shadow).** Means total one-side control. |
| **Engulfing** | One big candle **engulfs (swallows)** the previous smaller one. Hook: *big fish eats small fish.* |
| **Harami** | Japanese for *"pregnant."* A small candle sits inside the prior big candle's body — looks like a **pregnant belly** (big = mother, small = baby). |
| **Morning Star** | The "morning star" is the planet **Venus, which appears just before sunrise.** It signals **dawn = a new day = a bottom forming = bullish.** Hook: *morning = sun rising = price rising.* |
| **Evening Star** | The "evening star" is Venus **at dusk, before night falls.** Signals **darkness coming = a top = bearish.** Hook: *evening = sun setting = price setting.* |
| **Three White Soldiers** | Three strong green candles **marching upward like soldiers advancing** = strong bullish. |
| **Three Black Crows** | Three red candles stepping down — **crows = a bad omen**, descending = bearish. |
| **Tweezer Top/Bottom** | Two candles with **matching highs (or lows)** — like the two equal tips of a **pair of tweezers.** |

**Big picture:** candlesticks were invented by Japanese rice trader **Munehisa Homma** in the 1700s. The names are visual metaphors — so when you hear one, **picture the shape** and the meaning follows.

### Chart patterns (also mostly "it looks like...")

| Name | Memory hook |
|---|---|
| **Head & Shoulders** | The chart literally looks like a **head with a shoulder on each side.** Break the "neckline" = reversal. |
| **Double Top (M) / Double Bottom (W)** | The shape draws the letter **M** (two peaks = top) or **W** (two troughs = bottom). |
| **Cup & Handle** | Looks like a **teacup** (rounded bottom) with a small **handle** (dip) on the right, then breaks out. |
| **Triangle** | The converging trendlines form a **triangle.** |
| **Flag / Pennant** | A sharp move = the **flagpole**, then a small rectangle = the **flag** (or a tiny triangle = the **pennant**). |
| **Wedge** | Two lines tilting and converging like a **wedge** (doorstop) shape. |

## Indicators

| Name | What the name literally means (so you remember what it does) |
|---|---|
| **RSI — Relative Strength Index** | Measures the **strength of gains RELATIVE to losses.** The name = the formula: average gain ÷ average loss. |
| **MACD — Moving Average Convergence Divergence** | It tracks how two moving averages **converge (come together) and diverge (spread apart).** The name *is* what it computes. |
| **Bollinger Bands** | Named after the man who invented them — **John Bollinger** (1980s). "Bands" = the upper/lower lines. Hook: *a person's name.* |
| **Fibonacci** | Named after **Leonardo Fibonacci**, the medieval Italian mathematician whose number sequence (1,1,2,3,5,8,13…) produces the 61.8% "golden ratio." |
| **Stochastic** | From Greek *stokhos* = *"aim/guess"* — in statistics "stochastic" means **probabilistic.** It estimates where the close sits in its range. |
| **ADX — Average Directional Index** | An **average** of **directional** movement = an **index** of trend strength. |
| **ATR — Average True Range** | The **average** of the **"True Range"** (the full daily range including overnight gaps). |
| **OBV — On-Balance Volume** | Keeps a running **balance** of volume (adds on up days, subtracts on down days). |
| **VWAP — Volume-Weighted Average Price** | Exactly what it says: the average price **weighted by volume.** |
| **Ichimoku** | Japanese *"Ichimoku Kinko Hyo"* = *"one-glance equilibrium chart"* — see the whole balance **at one glance.** The shaded area = *Kumo* = "cloud." |
| **Pivot Point** | A **"pivot"** is the point something **turns/balances** on — the level price pivots around for the day. |
| **Golden Cross / Death Cross** | **Golden** = precious/good = bullish; **Death** = bad = bearish. The two MAs **cross.** |
| **Parabolic SAR** | **SAR = "Stop And Reverse"**; the dots trace a **parabola** (curve). |

## Theories named after people

| Name | The person |
|---|---|
| **Dow Theory** | **Charles Dow** — co-founder of the Wall Street Journal & Dow Jones index. |
| **Elliott Wave** | **Ralph Nelson Elliott** — accountant who mapped market "waves" in the 1930s. |
| **Wyckoff Method** | **Richard Wyckoff** — early-1900s trader who studied "smart money." |
| **Gann** | **W.D. Gann** — trader famous for angles/time analysis. |

## Options & derivatives (the BEST stories for memory)

| Name | Why it's named that |
|---|---|
| **Call option** | The right to **"call" the asset to you** — i.e., demand it / buy it. Hook: *call it IN = buy.* |
| **Put option** | The right to **"put" the asset onto someone** — i.e., sell/deliver it to them. Hook: *put it AWAY = sell.* |
| **Straddle** | You **"straddle" both sides** like sitting on a fence with one leg each side — buy a **call AND a put at the same strike.** Hook: *one leg each side = call + put.* |
| **Strangle** | Like a straddle but **hands spread wider apart** — the call and put are at **different (OTM) strikes.** Hook: *strangle = wider grip.* |
| **Covered Call** | The call you sold is **"covered"** because you **own the stock** to deliver. (If you don't own it, it's a risky **"naked"** call.) |
| **Iron Condor** | The profit zone is a **wide flat plateau with two "wings"** — like the broad wings of a **condor** (big bird). **"Iron"** = built from both calls and puts (a constructed, defined-risk frame). |
| **Butterfly** | The payoff diagram looks like a **butterfly** — a peak in the middle with a wing on each side. |
| **Contango** | Futures priced **higher** further out. Hook: *"Contango — Costs more, climbing up the curve."* (From an old London term for a fee to **continue/defer** settlement.) |
| **Backwardation** | Futures priced **lower** further out — the curve runs **"backward."** Hook: *backward = lower later.* |
| **The Greeks (Delta, Gamma, Theta, Vega)** | Greek letters for option sensitivities. **Delta** = change vs price. **Theta** = **T**ime decay. **Vega** = **V**olatility (note: Vega isn't a real Greek letter — it was picked because **V = Volatility**). |
| **VIX** | **V**olatility **I**nde**X**. Hook: *V for Volatility = the fear gauge.* |
| **Max Pain** | The strike that causes the **maximum pain (loss)** to the most option **buyers** at expiry. |
| **PCR — Put-Call Ratio** | Simply **puts ÷ calls** (open interest) — a sentiment gauge. |

# PART 15 — Formulas & Worked Examples

> You don't need to compute these by hand in the job (software does it). But knowing the formula lets you **explain what the number really means** — which is what an interviewer tests.

## Moving Averages

**SMA (Simple Moving Average)** = just the plain average of the last *n* closes.

```
SMA(n) = (P1 + P2 + ... + Pn) / n
```

*Example:* last 5 closes = 10, 12, 14, 16, 18 → SMA(5) = 70 / 5 = **14**.

**EMA (Exponential Moving Average)** = weights recent prices more (reacts faster).

```
k = 2 / (n + 1)                        (the "smoothing factor")
EMA_today = Price_today × k + EMA_yesterday × (1 − k)
```

*Example:* n = 10 → k = 2/11 = 0.182. If yesterday's EMA = 100 and today's price = 110: EMA = 110×0.182 + 100×0.818 = 20.0 + 81.8 = **101.8**.

## RSI — Relative Strength Index (the one you asked for, step by step)

```
1. For each day, find the change vs the previous close.
2. Split into GAINS (up days) and LOSSES (down days, as positive numbers).
3. Average Gain  = average of gains over 14 days
   Average Loss  = average of losses over 14 days
4. RS (Relative Strength) = Average Gain / Average Loss
5. RSI = 100 − (100 / (1 + RS))
```

*Worked example:* over 14 days, **Average Gain = 1.0** and **Average Loss = 0.5**.

```
RS  = 1.0 / 0.5 = 2
RSI = 100 − (100 / (1 + 2)) = 100 − 33.3 = 66.7   →  leaning overbought-ish
```

**Sanity checks (this is the intuition to say in an interview):** - All up days, no losses → RS = ∞ → **RSI = 100**. - All down days, no gains → RS = 0 → **RSI = 0**. - Gains = losses → RS = 1 → **RSI = 50** (neutral). So **above 70 = buyers strongly dominated recently (overbought); below 30 = sellers dominated (oversold).**

## MACD

```
MACD line   = EMA(12) − EMA(26)
Signal line = EMA(9) of the MACD line
Histogram   = MACD line − Signal line
```

*Example:* EMA12 = 1020, EMA26 = 1000 → MACD = **+20**. If Signal = 15 → Histogram = **+5** (positive = bullish momentum; MACD above signal).

## Bollinger Bands

```
Middle band = SMA(20)
σ (std dev) = standard deviation of the last 20 closes
Upper band  = Middle + 2σ
Lower band  = Middle − 2σ
```

*Example:* SMA20 = 100, σ = 5 → Upper = **110**, Lower = **90**. (±2σ statistically contains ~95% of recent prices, so touches are "stretched.")

## ATR — Average True Range

```
True Range (TR) = the largest of these three:
   (a) High − Low
   (b) | High − Previous Close |
   (c) | Low  − Previous Close |
ATR = average of TR over 14 days
```

*Example:* High = 105, Low = 100, Previous Close = 102. TR = max( 105−100=5 , |105−102|=3 , |100−102|=2 ) = **5**. (Using the prev close catches overnight **gaps**, which High−Low alone would miss.)

## Stochastic Oscillator

```
%K = (Close − Lowest Low) / (Highest High − Lowest Low) × 100     (over n, e.g. 14)
%D = 3-period average of %K
```

*Example:* Close = 50, lowest low = 40, highest high = 60 → %K = (50−40)/(60−40)×100 = **50**. (Near 100 = closing at the top of its range = strong.)

## Pivot Points (intraday)

```
Pivot (P) = (High + Low + Close) / 3          ← of the PREVIOUS day
R1 = 2P − Low       S1 = 2P − High
R2 = P + (High − Low)    S2 = P − (High − Low)
```

*Example:* yesterday H = 110, L = 100, C = 105 → P = 105. R1 = 2×105 − 100 = **110**, S1 = 2×105 − 110 = **100**, R2 = **115**, S2 = **95**.

## Fibonacci Retracement

```
For an up-move from Low (L) to High (H):  range = H − L
Retracement level (x%) = H − (x% × range)
Key levels: 23.6%, 38.2%, 50%, 61.8%
```

*Example:* L = 100, H = 200 → range = 100. 38.2% level = 200 − 38.2 = **161.8** ; 61.8% level = 200 − 61.8 = **138.2**. (The ratios come from the Fibonacci sequence: 8/13 ≈ 0.618, etc.)

## VWAP & OBV

```
VWAP = Σ(Price × Volume) / Σ(Volume)     (cumulative through the day)
```

*Example:* 10 shares @100 and 20 shares @102 → (1000 + 2040) / 30 = **101.33**.

```
OBV:  if Close > prev Close → OBV = OBV + Volume
      if Close < prev Close → OBV = OBV − Volume
```

## Backtesting / performance metrics (your backtester project)

```
Daily return r    = (Price_today − Price_yesterday) / Price_yesterday
CAGR              = (Ending Value / Starting Value)^(1 / years) − 1
Sharpe ratio      = (mean daily return / std dev of daily returns) × √252
Sortino ratio     = (mean daily return / std dev of NEGATIVE returns only) × √252
Max Drawdown      = max over time of (Peak − Trough) / Peak
Profit Factor     = gross profit / gross loss
Win rate          = winning trades / total trades
```

*Worked examples:* - **CAGR:** 100 → 200 over 4 years = 2^(1/4) − 1 = 1.189 − 1 = **18.9% per year**. - **Sharpe:** mean daily 0.0008, std 0.01 → (0.0008/0.01) × √252 = 0.08 × 15.87 = **1.27** (good, >1). - **Max Drawdown:** equity goes 100 → 150 → 90. Peak = 150, trough = 90 → (150−90)/150 = **40%**. - **Profit Factor:** total winners +30, total losers −15 → 30/15 = **2.0** (made ₹2 for every ₹1 lost). - *(√252 annualises daily numbers because there are ~252 trading days a year.)*

## Options math

```
Intrinsic value (Call) = max(Spot − Strike, 0)
Intrinsic value (Put)  = max(Strike − Spot, 0)
Time value             = Premium − Intrinsic value
Long Call payoff@expiry = max(Spot − Strike, 0) − Premium
Long Put  payoff@expiry = max(Strike − Spot, 0) − Premium
Breakeven (long call)  = Strike + Premium
Breakeven (long put)   = Strike − Premium
PCR                    = Put Open Interest / Call Open Interest
```

*Example:* Buy a 24,000 call for ₹100 premium; at expiry Nifty = 24,250. Intrinsic = 24,250 − 24,000 = 250. Profit = 250 − 100 = **₹150 per unit**. Breakeven was 24,100. **Implied Volatility (IV):** there's no simple formula — it's **solved by reversing the Black-Scholes option-pricing model**: you know the option's market price and back out the volatility number that makes the model match. High IV = expensive options = market expects big moves.

---

### How to use this sheet

1. **Part 1** — read each story once; the name will trigger the meaning in the interview ("Morning Star = Venus before sunrise = bottom = bullish").
2. **Part 2** — you don't memorise to compute by hand; memorise enough to **explain** ("RSI is average gain over average loss, scaled 0–100; 70+ means buyers dominated and it's overbought").
3. The killer combo: when asked about RSI, give the **formula + the sanity check** (all gains = 100, all losses = 0, equal = 50). That proves you understand it, not just recite it.

# PART 16 — Interview Prep: Self-Intro, Rapid Revision & Q&A

### 0. Your 30-second self-intro (memorise the shape, not the words)

> "I'm an MBA-Finance candidate and NISM Series-XV Research Analyst certified. I have close to two years on a derivatives desk at D.E. Shaw as Member Technical, where I analysed F&O market data — implied volatility, open interest, option greeks — and prepared daily reports for the trading team. Alongside that I do technical analysis on Nifty, Bank Nifty and commodities, and I've built tools to backtest strategies and generate daily research reports. I'm certified to publish buy/sell research and I want to do exactly that as a Technical Research Analyst."

---

### PART A — What the job actually is

A **Technical Research Analyst (TRA)** studies **price charts** (not company balance sheets) to predict where Commodity / Futures & Options / equity prices are headed, and publishes **Buy / Sell / Hold calls** with a target and stop-loss in **daily, weekly and monthly reports**, and supports relationship managers (RMs) with market views.

**Technical analysis vs Fundamental analysis** (very common question): - **Fundamental** = *what to buy*. Studies company financials, earnings, valuation, economy. - **Technical** = *when to buy/sell*. Studies price & volume charts, assuming price already reflects all information and that trends and patterns tend to repeat.

**The 3 assumptions of technical analysis** (say these and you sound legit): 1. **Price discounts everything** — all news/info is already in the price. 2. **Prices move in trends** — once a trend starts it tends to continue. 3. **History repeats** — patterns recur because human behaviour (fear/greed) repeats.

---

### PART B — The indicators ON YOUR RESUME (you MUST be able to explain each)

#### 1. Trend

- **Uptrend** = higher highs + higher lows. **Downtrend** = lower highs + lower lows. **Sideways** = range.
- Golden rule: *"Trade with the trend."* You buy dips in an uptrend, sell rallies in a downtrend.

#### 2. Support & Resistance

- **Support** = a price *floor* where buyers repeatedly step in and stop the fall.
- **Resistance** = a price *ceiling* where sellers repeatedly step in and stop the rise.
- A **breakout** above resistance (or below support) signals a new move. Old resistance, once broken, often becomes new support (and vice-versa).
- *Interview line:* "I buy near support with a stop just below it, and target the next resistance."

#### 3. Moving Averages (MA / DMA — Daily Moving Average)

- Average of the last N closing prices; smooths out noise to show the trend.
- **SMA** = simple average. **EMA** = exponential, weights recent prices more (reacts faster).
- Common ones: **20-DMA** (short term), **50-DMA** (medium), **200-DMA** (long term).
- Price **above** its MA = bullish; **below** = bearish.
- **Golden Cross** = 50-DMA crosses *above* 200-DMA → bullish. **Death Cross** = opposite → bearish.

#### 4. RSI — Relative Strength Index

- A momentum oscillator from **0 to 100** measuring how strong recent gains are vs losses (default 14 days).
- **Above 70 = overbought** (may pull back). **Below 30 = oversold** (may bounce).
- **Divergence** = price makes a new high but RSI doesn't → momentum weakening, possible reversal.
- *Interview line:* "RSI tells me if a move is overstretched; I don't chase a stock at RSI 80."

#### 5. MACD — Moving Average Convergence Divergence

- = (12-day EMA − 26-day EMA), plus a **9-day signal line**, plus a **histogram** of the gap.
- **Bullish** when the MACD line crosses *above* the signal line; **bearish** when it crosses below.

- Tells you momentum and trend direction together.

### 6. Bollinger Bands

- A 20-day moving average with an upper and lower band at **±2 standard deviations**.
- They measure **volatility**: bands widen when volatile, narrow ("squeeze") when calm.
- Price near the **upper band** = strong/overbought; near the **lower band** = weak/oversold.
- A **squeeze** (very narrow bands) often comes *before* a big breakout.

### 7. Volume

- Number of shares/contracts traded. **Volume confirms price.** A breakout on high volume is reliable; on low volume it's suspect.

### 8. Candlestick patterns (each candle shows Open, High, Low, Close)

- Green/white = close above open (up day). Red/black = close below open (down day).
- **Body** = open-to-close; **wicks/shadows** = the highs/lows.
- **Doji** = tiny body → indecision, possible reversal.
- **Hammer** = small body at top + long lower wick → bullish reversal after a fall.
- **Bullish Engulfing** = a big green candle fully covers the prior red candle → bullish reversal.
- **Shooting Star** = small body at bottom + long upper wick → bearish reversal.

### 9. Chart patterns (know the names + what they signal)

- **Head & Shoulders** = reversal (top = bearish). **Double Top / Bottom** = reversal.
- **Triangles / Flags / Pennants** = continuation (trend resumes after a pause).

### 10. Fibonacci retracement

- After a move, prices often pull back to **23.6%, 38.2%, 50%, 61.8%** of that move before continuing.
- These levels act as support/resistance. **61.8%** is the "golden ratio" and the most watched.

### 11. ATR — Average True Range

- Measures **how much an asset typically moves in a day** (volatility). Used to **size stop-losses**: a wider ATR → wider stop. *Interview line:* "I set my stop ~1.5×ATR away so normal noise doesn't hit it."

---

## PART C — How to make a Buy/Sell call (the analyst's core job)

Every call has **4 parts**: 1. **View**: Buy / Sell / Hold. 2. **Entry**: the price to act at. 3. **Target**: where you book profit (often the next resistance, or a risk:reward multiple). 4. **Stop-loss**: where you admit you're wrong and exit (protects capital).

- **Risk:Reward** = (target − entry) ÷ (entry − stop). Aim for **at least 1:1.5 or 1:2**.
- *Example to say:* "Buy Bank Nifty at 58,400, target 60,300, stop 57,300 — that's roughly 1:1.7 reward-to-risk, and I'd exit if it closes below the 50-DMA."
- **Golden rule:** a good analyst is disciplined about **stop-losses**. "I'd rather take a small loss than a big one." Never average down a losing trade without a reason.

---

## PART D — The markets you'll be asked about

### Nifty 50 & Bank Nifty

- **Nifty 50** = index of the 50 largest NSE companies → the benchmark for the Indian market.
- **Bank Nifty** = index of the 12 largest banks → **more volatile**, moves faster, heavily traded in F&O.
- **Sensex** = BSE's 30-stock equivalent of Nifty.

### Futures & Options (F&O / Derivatives) — you worked on this desk

- A **derivative** derives its value from an underlying (stock/index/commodity).
- **Futures** = an *obligation* to buy/sell at a set price on a future date.
- **Options** = a *right* (not obligation):

- **Call option** = right to **buy** (you buy calls if you're bullish).
- **Put option** = right to **sell** (you buy puts if you're bearish).
- **Strike price** = the agreed price. **Premium** = what you pay for the option. **Expiry** = last day.
- **Open Interest (OI)** = total open contracts; rising OI + rising price = strong trend.
- **Implied Volatility (IV)** = the market's expectation of future volatility, priced into options. High IV = expensive options. (You analysed IV surfaces — be ready to say "IV across strikes/expiries".)
- **Option Greeks** (you tracked these):
- **Delta** = how much the option moves per ₹1 move in the underlying.
- **Gamma** = how fast delta changes.
- **Theta** = time decay (options lose value as expiry nears).
- **Vega** = sensitivity to volatility.
- **Strategies**: **Straddle** (buy call+put same strike — bet on a big move either way), **Strangle** (same idea, different strikes, cheaper), **Spread** (buy+sell options to cap risk/cost).

### Commodities (MCX) — the JD's first bullet

- Traded on **MCX** (Multi Commodity Exchange) in India: **Gold, Silver, Crude Oil**, natural gas, copper.
- Indian prices track **global benchmarks**: Gold → COMEX (USD/ounce), Crude → WTI/Brent (USD/barrel).
- **Drivers**: the **US dollar** (stronger USD → weaker gold), **interest rates & inflation** (gold is an inflation hedge / safe haven), **geopolitics & supply** (crude reacts to OPEC, wars), and **demand**.
- Same technical tools (S/R, RSI, MACD, MAs) apply to commodity charts.

---

### PART E — NISM & the rules (you're certified — know the basics)

- **NISM Series-XV: Research Analyst** is the SEBI-mandated certification to legally publish research / buy-sell recommendations in India.
- Key principles a Research Analyst must follow:
- **Disclosure** of any holding/conflict of interest in a stock you recommend.
- **No front-running** — you can't trade ahead of your own published call.
- **Separation** of research from the dealing/trading function.
- Recommendations must have a **rational basis** and be documented.
- *Interview line:* "Being NISM-certified means I can publish compliant research with proper disclosures and a documented rationale."

---

### PART F — How to explain YOUR 3 projects (they WILL ask)

#### 1. Daily/Weekly/Monthly Technical Research Reports (Nifty, Bank Nifty, Gold, Crude):

> "I prepare a morning note on indices and commodities — I mark support/resistance, check RSI, MACD, Bollinger Bands and candlestick patterns, and give a Buy/Sell/Hold call with entry, target and an ATR-based stop-loss. I automated the data part in Python so the levels stay current."

#### 2. Multi-Strategy Technical Backtester (Python):

> "I backtested four technical strategies — MA crossover, RSI mean-reversion, MACD trend and Bollinger breakout — on six years of Nifty and Bank Nifty data, and compared them to buy-and-hold on CAGR, Sharpe, Sortino, max drawdown, win-rate and profit factor. The active strategies underperformed buy-and-hold in a strong bull market, but **MACD-trend gave the best risk-adjusted return with the smallest drawdown** — the real lesson was validating strategies with data instead of trusting them blindly." *(This honesty + the metrics vocabulary impresses interviewers.)*

**Backtesting metrics — be able to define these (they're in your report):** - **CAGR** = the smoothed annual growth rate of the strategy (return per year). - **Sharpe ratio** = return per unit of *total* risk (volatility). Higher is better; >1 is good. - **Sortino ratio** = like Sharpe but only penalises *downside* volatility — a fairer risk measure. - **Max Drawdown** = the worst peak-to-trough fall the strategy suffered (how much pain you'd endure). - **Win-rate** = % of trades that were profitable. - **Profit Factor** =

gross profit ÷ gross loss; above 1 means the system made money overall. - **Exposure** = % of the time the strategy was actually invested (rest in cash).

**3. Multi-Asset Technical Coverage Watchlist:**

> "A ranked watchlist of 10 NSE stocks plus Gold and Crude, scored on trend, momentum and candlestick signals, tagging Buy/Sell/Hold with stop-losses — the kind of coverage sheet you'd hand to RMs each morning."

---

### PART G — 20 likely questions (one-line answers to expand on)

1. **What is technical analysis?** → Studying price/volume charts to forecast prices; price discounts everything.
2. **Technical vs fundamental?** → Fundamental = what to buy; technical = when to buy/sell.
3. **What is support & resistance?** → Floor where buyers step in / ceiling where sellers step in.
4. **Explain RSI.** → 0–100 momentum; >70 overbought, <30 oversold.
5. **Explain MACD.** → 12-EMA minus 26-EMA with a 9-signal line; cross above = bullish.
6. **What are moving averages? Golden cross?** → Smoothed avg price; 50 crossing above 200 = bullish.
7. **Bollinger Bands?** → 20-MA ±2 SD; measures volatility; squeeze precedes breakout.
8. **What's a candlestick? Name a reversal pattern.** → OHLC bar; hammer / bullish engulfing.
9. **Fibonacci retracement levels?** → 23.6/38.2/50/61.8%; pullback support/resistance.
10. **How do you decide a buy?** → Trend up + price above MAs + RSI not overbought + MACD bullish + near support.
11. **What is a stop-loss and why?** → A pre-set exit to cap losses; protects capital and emotions.
12. **Risk-reward ratio?** → Reward ÷ risk; want ≥1:2.
13. **Difference between Nifty and Bank Nifty?** → 50-stock benchmark vs 12 banks, Bank Nifty more volatile.
14. **Call vs Put option?** → Right to buy vs right to sell.
15. **What is Open Interest / Implied Volatility?** → Open contracts / market's expected future volatility.
16. **Name the option greeks.** → Delta, Gamma, Theta, Vega.
17. **What's a straddle?** → Buy call + put at same strike to profit from a big move either way.
18. **What drives gold / crude prices?** → USD, rates, inflation, safe-haven / OPEC, supply-demand, geopolitics.
19. **What does NISM RA certification allow?** → To publish compliant research with disclosures.
20. **Walk me through one of your buy calls.** → Use the Bank Nifty example in Part C.

---

### PART H — Smart questions to ASK them (shows interest)

- "Which segments will I cover — equity, F&O, or commodities — and on what timeframes?"
- "Do analysts here publish their own calls or support senior analysts first?"
- "What tools/terminals does the desk use — TradingView, broker terminal, Bloomberg?"
- "How is a call's performance tracked here?"

### Dos & Don'ts

- ✅ Speak in terms of **trend, levels, risk-reward, stop-loss** — that's the analyst's language.
- ✅ Admit what you don't know briefly, then pivot to what you do.
- ✅ Always attach a **stop-loss** to any call you mention — never sound reckless.
- ❌ Don't claim you made live buy/sell calls at D.E. Shaw — you were **Member Technical** (analysis/support).
- ❌ Don't guarantee returns or say "this will definitely go up." Analysts talk in probabilities and risk.

---

### Tonight's priority order (if short on time)

1. Part B (indicators) + Part C (buy/sell call) — **non-negotiable**, it's your core.
2. Part A (TA vs fundamental, 3 assumptions) — easy marks.
3. Part D (Nifty/Bank Nifty, options basics, commodity drivers).
4. Part F (your projects) + Part G (Q&A) — rehearse out loud.
5. Part E (NISM) + Part H — quick skim.

```
Good luck. You know more than you think once you read this twice.
```